National Environment Agency

Celebrating 20 Years of Environmental Stewardship




# Championing ENVIRONMENTAL STEWARDSHIP for a Sustainable Future

## Integrated Sustainability Report 2021/2022

# Contents


**Vision, Mission, Roles and Core Values** 3

**Chairman's Foreword** 4

**CEO's Message** 5

**Corporate Governance** 6

- BOARD OF DIRECTORS
- SENIOR LEADERSHIP TEAM

**About the National Environment Agency** 9

- VALUE CREATION MODEL
- MATERIAL ISSUES
- RISKS AND OPPORTUNITIES

**Chapter 1** 18
**Ensuring a Clean and Sustainable Environment for Singapore**

- HIGH STANDARDS OF PUBLIC HEALTH
- SAFE, HEALTHY AND CONDUCIVE LIVING ENVIRONMENT
- SUSTAINABLE AND RESOURCE EFFICIENT SINGAPORE
- PROVIDING TIMELY AND RELIABLE WEATHER AND CLIMATE SERVICES
- VIBRANT HAWKER CULTURE

**Chapter 2** 30
**Together with Our Partners and the Community**

- ENVIRONMENTAL SERVICES INDUSTRY AS A FORCE MULTIPLIER FOR MISSION SUCCESS
- ACTIVE COMMUNITY AS STEWARDS OF THE ENVIRONMENT
- PUBLIC SECTOR TAKING THE LEAD IN ENVIRONMENTAL SUSTAINABILITY
- INTERNATIONAL PARTNERSHIPS AS STRATEGIC LEVERS

**Chapter3** 35
**Enabled by a High Performance and Future-ready NEA**

- CREATION OF CAPACITY
- STRONG CAPABILITIES FOR MISSION SUCCESS
- STRONG CORPORATE PRACTICES FOR A SUSTAINABLE ORGANISATION
- CONDUCIVE CULTURE FOR COLLABORATION, LEARNING AND INNOVATION
- WALKING THE TALK

**About This Report** 49

- INTEGRATED REPORTING APPROACH

**GRI Content Index** 51

# Vision

A clean environment, towards a liveable and sustainable Singapore

# Mission

To ensure a clean and sustainable environment for Singapore, together with our partners and the community

# Roles

- Maintain high standards of public health
- Improve and sustain a clean environment
- Provide up-to-date and reliable weather information services
- Promote resource efficiency and conservation in collaboration with our partners and the community

# Core Values

**Care**
NEA values each individual and strives to exceed our customers' expectations

**Integrity**
NEA upholds the trust and confidence of others in us

**Teamwork**
NEA works as a team and with our partners to achieve common goals

**Innovation**
NEA constantly pursues new ideas and creative solutions

**Professionalism**
NEA takes pride in what we do and strives for excellence

# Chairman's Foreword



**Mr Lee Chuan Seng**
**Chairman**

2022 is a special year for NEA, as we celebrate our 20th Anniversary. As the leading public organisation responsible for a clean and liveable Singapore, sustainability has been central to NEA's mandate since our formation in July 2002.

And today, faced with a multitude of environmental threats, NEA adopts a multi-pronged approach by combining policy changes with stakeholder engagement to encourage sustainable production, consumption and development. Like most sustainable cities, Singapore strives to become a Zero Waste Nation by embracing circular economy across the value chain. To accelerate the adoption of waste and resource management initiatives, NEA administered a $45 million 'Closing the Waste Loop' fund to develop innovative solutions that could possibly extend the lifespan of Semakau Landfill.

NEA also plays a vital role in our nation's efforts to strengthen climate resilience. To meet international obligations to the Kigali Amendment, NEA has introduced measures to reduce the usage and production of hydrofluorocarbons which are potent greenhouse gases (GHG), and shift the industry towards climate-friendly technologies.

Beyond carbon mitigation, NEA also supports Singapore's vision to become a leading centre for green finance through the launch of the Multicurrency Medium Term Note Programme and Green Bond Framework last year, a first by a statutory board in Singapore. The proceeds from the issuance of Notes will go towards funding sustainable infrastructure development projects.

Amid the uncertainty of the pandemic, we held our first hybrid format CleanEnviro Summit Singapore Catalyst in September 2021, facilitating global discourse on sustainable development. The event was a successful lead-up to the flagship CleanEnviro Summit Singapore in April 2022, providing opportunities for public and private sector leaders from around the world to convene and address climate risk and urban waste issues ahead of the 27th United Nations (UN) Climate Change conference.

Looking back, NEA has come a long way in the past two decades. Although the challenges we face today are vastly different, we will forge ahead to create value for our stakeholders, as well as catalyse and support industry-wide, sustainable change.

NEA will continue to play a vital role in safeguarding our environment for future generations, in the next 20 years and beyond.

# CEO's Message



**Mr Luke Goh**
**Chief Executive Officer**

2022 marks NEA's 20th year of stewardship and safeguarding the environment. The past year has been our most exciting and challenging yet, as we engaged numerous stakeholders on sustainability initiatives while supporting the nation's transition into COVID-19 endemicity.

The COVID-19 pandemic has underscored the importance of upholding high public hygiene standards in Singapore. NEA has progressively rolled out the Environmental Sanitation Regime to mandate baseline hygiene and sanitation standards for high-risk premises, and received strong support from the community to implement tray and crockery return practices in hawker centres, coffeeshops and similar settings. At the peak of the pandemic, NEA undertook the implementation of safe management measures at hawker centres, supported the mandatory COVID-19 routine regular testing for stallholders, cleaners and related sectors, and ensured adequate capacity at all our after-death facilities.

NEA is pressing on with initiatives to reduce waste, encourage recycling, and ultimately lower carbon emissions. To encourage more sustainable habits, a disposable carrier bag charge will be implemented at supermarkets in 2023. Meanwhile, a beverage container return scheme is being developed where manufacturers and importers of pre-packaged beverages will refund consumers a modest deposit to encourage the return and recycling of empty beverage containers. NEA is also exploring chemical recycling solutions to help close the plastic waste loop in Singapore.

In the area of weather and climate science, NEA's Meteorological Service Singapore continues to provide round-the-clock weather services to the aviation and shipping sectors, as well as the public. To build Singapore's climate resilience, we embarked on the Climate Science Research Masterplan to deepen our understanding of climate change parameters. We also pressed on with international scientific collaborations, and became a core member in the international Unified Model Partnership comprising Australia, India, New Zealand, the Republic of Korea, and the United Kingdom.

Meanwhile, we continue our efforts against public health threats such as dengue. Despite a significant decrease in dengue cases in 2021, there was a resurgence of cases at the start of 2022 signalling early signs of an outbreak. NEA and our partners rapidly strengthened vector control measures in tandem with the steady expansion of Project *Wolbachia* to bring the wave of cases under control.

Today's NEA is defined by the sacrifices and forward planning of our predecessors in the last twenty years. To weather the uncertainties of tomorrow, today's management and staff are likewise planning for the next twenty years – setting for ourselves aspirations to extend the lifespan of Semakau Landfill almost indefinitely, expanding our network of hawker centres anchored on the heritage of a thriving hawker culture, growing new capabilities to bring recycling and public health outcomes to fresh heights.

As we celebrate NEA's 20th Anniversary, we remain mindful of our role as stewards, entrusted with the responsibility of safeguarding our environment for future generations. We call on all stakeholders and friends to join us in this continuing endeavour, pressing on with gusto and gumption!

# Corporate Governance

## BOARD

The NEA Board provides strategic direction and oversight of organisational functions and goals, and steers NEA in fulfilling our vision and mission. In the Financial Year (FY) 2021, the Board comprised 14 members with diverse and extensive experience from both the private and public sectors.

Appointed by the Minister for Sustainability and the Environment, the Board is guided by a set of key principles to ensure fairness and impartiality in carrying out its responsibilities, as well as to uphold the integrity and reputation of the Public Service. Led by Chairman Mr Lee Chuan Seng, the Board has set up committees to assist with the execution of its responsibilities.

The Audit Committee assists the Board in maintaining a high standard of corporate governance, particularly in the areas of financial accounting and reporting, risk management, and internal control systems. Chaired by Ms Jackie Chew, the Audit Committee comprises Ms Sylvia Choo, Mr Puah Kok Keong, Dr Andrew Khoo, and Mr Bernard Choo. The Committee reviews the annual accounts, audit plans, internal control systems, as well as risk and audit reports with NEA's external and internal auditors.

The Human Resource Committee assists the Board in ensuring an objective and fair performance management process for NEA. Chaired by Mr Lee Chuan Seng, the Human Resource Committee comprises Deputy Chairman Mr Ravinder Singh, Ms Lynette Leong (till 31 March 2022), Mr Lim Eng Hwee, Mr Zia Zaman (as of 1 April 2022), and NEA's Chief Executive Officer (CEO).

The Finance Committee oversees matters related to the investment of surplus funds, cash and working capital management, and fundraising activities. Chaired by Mr Quek Suan Kiat, the Finance Committee comprises Mr Suhaimi Zainul-Abidin, Ms Susan Chong, Professor Ho Teck Hua (till 31 March 2022), Ms Janet Young (as of 1 April 2022), and NEA's CEO.

## SENIOR MANAGEMENT

NEA's senior management[1], led by our CEO, is responsible for implementing the strategic direction of NEA's programmes, and nurturing a high-performance organisation committed to fulfilling NEA's environmental mission and roles. They manage the financial, manpower and operational resources of their respective divisions and departments, and work together at committee meetings to guide NEA's continued performance in risk management, innovation and sustainability practices.

The Senior Leadership Team meets regularly to direct and review management and operational policies and activities. It also leads and directs NEA's strategies and transformational efforts. Through strategically-themed meetings and projects, the Senior Leadership Team guides NEA in becoming more effective in our dual roles as regulator and champion for the environment. These efforts are aimed at improving how NEA is organised, leveraging technology and utilising resources, thereby enabling better results by working smarter and striving for organisational excellence.

The pay structure for senior management consists of fixed pay and variable pay. While the variable pay component is influenced by economic factors, it is directly linked to the individual's performance achievement. The ratio of variable pay to fixed pay for senior management is relatively higher as this drives performance, which is reviewed and assessed by the Human Resource Committee. The Committee provides an objective and fair performance assessment, and approves the corresponding recommendation of performance bonuses and salary increments.

## CODE OF CONDUCT

NEA has an internal code of conduct that applies to all employees as NEA has a zero-tolerance policy on fraud and misconduct by employees. Called the NEA Person Code, the code of conduct articulates NEA's vision, mission and values, appropriate conduct, and guidelines concerning conflicts of interest, confidentiality, corruption prevention and proper financial stewardship.

The Code also sets out principles to guide employees in discharging their duties and responsibilities to the highest standards of personal and corporate integrity. The NEA Person Code is readily available on the NEA intranet and in the Staff Handbook.

All new staff are informed about the NEA Person Code. Reminders on the code are also regularly sent to staff. In the event of a breach of discipline, the NEA Disciplinary Policy details the appropriate procedures to deal with the breach. Under NEA's whistleblowing policy, employees and third parties are encouraged to report any incidents of fraud and serious misconduct. NEA also offers employees ample channels to air their grievances on matters that affect the terms and conditions of their service. They are reminded to declare as and when there is an actual or potential conflict of interest in their official duties and personal interests, during the course of their work. Should there be declarations, the relevant parties would be informed.

## AUDIT

Each year, NEA's accounts are audited by external auditors, approved by the Board and published with our Integrated Sustainability Report. The external auditors are appointed by the Minister for Sustainability and the Environment for a fixed term, in consultation with the Auditor-General. In addition, the Auditor-General's Office and Accountant-General's Department also perform both scheduled and ad hoc audits to ensure that NEA continues to adhere to good governance practices.

The external auditors provide the Board with an independent opinion on NEA's financial statements, offering reasonable assurance that the financial statements are free from material misstatements. They conduct the audit in accordance with the audit plan presented to the Audit Committee. The external auditors are required to declare their state of independence to the Audit Committee prior to their appointment. Significant non-audit services to be provided by the external auditors are also subjected to approval by the Audit Committee.

NEA's Internal Audit Division provides independent assurance and advisory services to management and the Audit Committee. Reviews performed by the Internal Audit Division assist the Board in promoting good corporate governance by assessing the design and operating effectiveness of controls. The reviews also focus on compliance with NEA's policies and procedures.

[1] *For NEA's organisation chart, visit www.nea.gov.sg/corporate-functions/who-we-are/people/organisation-chart*

# BOARD OF DIRECTORS



**Mr Lee Chuan Seng**

Chairman, National Environment Agency

Emeritus Chairman, Beca Asia Holdings Pte Ltd



**Mr Ravinder Singh**

Deputy Chairman, National Environment Agency

Group Chief Operating Officer, Technology & Innovation

President, Defence & Public Security, Singapore Technologies Engineering Ltd



**Mr Luke Goh**

Chief Executive Officer, National Environment Agency



**Ms Jackie Chew**

Chief Risk Officer, Prudential Assurance Company Singapore (Pte) Ltd



**Mr Bernard Choo**

Editor, Shin Min Daily News, Singapore Press Holdings Media Trust



**Mr Quek Suan Kiat**

Former Vice Chairman, Barclays Bank PLC Singapore



**Ms Sylvia Choo**

Director, Unions, National Trades Union Congress and Co-General Manager, Tripartite Alliance for Dispute Management



**Mr Suhaimi Zainul-Abidin**

Chief Executive Officer, Quantedge Capital Pte Ltd



**Mr Lim Eng Hwee**

Chief Executive Officer, Urban Redevelopment Authority



**Mr Puah Kok Keong**

Deputy Secretary (Policy), Ministry of Home Affairs



**Ms Susan Chong**

Chief Executive Officer, Greenpac (S) Pte Ltd



**Dr Andrew Khoo**

Former Deputy Managing Director, Monetary Authority of Singapore



**Ms Lynette Leong**

Chief Sustainability Officer, CapitaLand Investment

*(till 31 March 2022)*



**Professor Ho Teck Hua**

Senior Deputy President and Provost, National University of Singapore

*(till 31 March 2022)*



**Ms Janet Young**

Managing Director & Group Head, Channels & Digitalisation,

Strategic Communications & Brand, United Overseas Bank Ltd

*(with effect from 1 April 2022)*



**Mr Zia Zaman**

General Manager, Corporate Business Development Asia, Microsoft

*(with effect from 1 April 2022)*

*As of 1 April 2022*

# SENIOR LEADERSHIP TEAM



**Mr Luke Goh**
Chief Executive Officer



**Mr Ananda Ram Bhaskar**
Deputy Chief Executive Officer, Environmental Protection
Director-General, Environmental Protection



**Mr Ng Chun Pin**
Deputy Chief Executive Officer, Planning, Corporate & Technology



**Mr Chew Ming Fai**
Deputy Chief Executive Officer, Public Health
Director-General, Public Health



**Ms Koh Li-Na**
Assistant Chief Executive, Meteorological Services and Radiation Protection



**Ms Wong Chin Ling**
Director-General, Meteorological Services



**Mr Khoo Seow Poh**
Senior Advisor



**Mr Cheang Kok Chung**
Deputy Director-General, Environmental Protection
Group Director, Resource & Sustainability



**Ms Tan Li Yen**
Group Director, Clean Environment



**Mr Tan Eng Kim Francis**
Group Director, Corporate Services



**Dr Ng Lee Ching**
Group Director, Environmental Health Institute



**Mr Tony Teo**
Group Director, Environmental Public Health Operations



**Mr Andrew Low**
Group Director, Hawker Centres



**Mr Desmond Tan**
Group Director, Infrastructure & Engineering



**Mr S Satish Appoo**
Group Director, Joint Operations & Technology



**Mr Lim Yuin Chien**
Group Director, Public Engagement



**Mr Fong Peng Keong**
Group Director, Public Health Services & Capabilities



**Mr Ang Kok Kiat**
Group Director, Radiation Protection and Nuclear Science



**Ms Ivy Ong Bee Luan**
Group Director, Strategic Planning & Policy

*As of 1 September 2022*

# About the National Environment Agency

NEA was established on 1 July 2002 by the NEA Act 2002 as a statutory board under the Ministry of Sustainability and the Environment of Singapore.

NEA is a member of these international sustainability associations:

- United Nations Global Compact (UNGC), since September 2016
- Global Compact Network Singapore, a local chapter of UNGC, since February 2010

As regulator and advocate for the environment, NEA strives to formulate clear policies and public initiatives that benefit Singapore's environment, economy and society in the long run.

The following diagram outlines our key strategies to achieve our vision of a clean environment, towards a liveable and sustainable Singapore.



# VALUE CREATION MODEL

NEA transforms our inputs, through our business activities, into outputs and outcomes for NEA's five key roles in public health, environmental protection, resource sustainability, weather and climate services, and hawker culture. To achieve these outcomes, NEA works closely with our partners and the community, as well as develops our capabilities and human capital to be a high performance and future-ready organisation.

## OUR MISSION

NEA's mission is to ensure a clean and sustainable environment for Singapore, together with our partners and the community. Our mandate is aligned with the Singapore Agenda on achieving a Sustainable, Resource Efficient and Climate Resilient Singapore.

## WHAT WE DO

We strive to achieve our mission through five key roles



### High Public Health Standards

NEA maintains an effective surveillance programme for vector-borne diseases and the control of vector populations. We shape social norms against littering and smoking in prohibited places, and maintain an effective cleaning regime in public places. To guard against environment-related diseases, we maintain high environmental hygiene standards. Besides planning the supply of after-death facilities, NEA also operates Singapore's public after-death facilities, providing cremation, burial, columbarium and ash scattering services.



### Safe, Healthy and Conducive Living Environment

We monitor and prevent air and water pollution, and regulate hazardous substances and toxic industrial waste. We ensure a quality living environment through upstream land use planning and siting of industries, and implementation of noise management measures. NEA also safeguards the public, radiation workers, and the environment against the harmful effects of radiation.



### Sustainable and Resource Efficient Singapore

NEA promotes energy efficiency and conservation to improve air quality, reduce GHG emissions, and mitigate climate change. We plan, develop and manage the solid waste management system in Singapore, and strive towards a Zero Waste Nation by promoting circular economy and 3R (reduce, reuse and recycle) practices to conserve resources.



### Timely and Reliable Weather and Climate Services

NEA provides reliable and responsive weather and climate services, disseminating weather forecasts and hazard warnings in a timely manner and contributing to climate adaptation efforts. We provide fit-for-purpose weather information and climate services to support public safety and socio-economic activities. We also issue haze alerts and provide vital meteorological services to the aviation and maritime communities.



### Vibrant Hawker Culture

We develop and manage hawker centres as vibrant communal spaces, offering a wide variety of affordable food, in a clean and hygienic environment. NEA also ensures the sustainability of the hawker trade, by facilitating training and skills development for both aspiring and existing hawkers, and ensuring decent livelihoods of hawkers.

## WHAT WE RELY ON

We draw upon six capitals as part of our organisational strategy and business model



### Working Together with Partners and Community

Collaborating and co-creating with our partners and the community to bring a greater positive impact on the environment



### Delivering Citizen-centric Services

Delivering citizen-centric services and being well-placed to take on public health threats together



### Creating Capacity

Strengthening key operational domains in NEA and reinventing service delivery and enforcement



### Building Strong Capabilities

Building strong capabilities to deliver on work areas and ensure preparedness to mitigate future challenges



### Developing and Engaging Staff

Developing a highly efficient and effective workforce by promoting a culture of lifelong learning and instilling a spirit of innovation in staff



### Strengthening Corporate Practices and Governance

Strengthening corporate practices and governance for a future-ready organisation

## THE VALUE WE CREATE

Sustainability is high on NEA's agenda. We consider the wider value we deliver against each capital

**92** per cent of respondents to the Public Cleanliness Satisfaction Survey were satisfied or very satisfied with overall public cleanliness

**96** local dengue (dengue fever and dengue haemorrhagic fever) cases per 100,000 population[2]

**99.5** per cent of days in a year where Pollution Standards Index is in the 'Good' and 'Moderate' range

**10** percentage points increase in overall recycling rate compared to 2002

**5** per cent reduction in domestic waste disposed of per capita from 2012

**91** per cent accuracy of Nowcast

**114** hawker centres and markets managed by NEA and NEA-appointed operators

[2] *The outbreak years of 2013 and 2014 had an incidence rate of 405 and 326 cases per 100,000 population*

# MATERIAL ISSUES

Our material issues are refined every three years, in consultation with the NEA Groups and Divisions, based on their work priorities and areas critical to mission success. We also reviewed and considered key trends and driving forces in the operating environment, all of which would have significant impact on NEA's work and operations. The stakeholders we engaged have worked collectively and collaboratively with NEA to make a positive impact on Singapore's environment.

- Employees
- Industry
- Non-governmental Organisations (NGOs)
- Schools
- Public
- Community
- Partners
- Public Sector Agencies

| Material Issues | Impact on NEA and Stakeholders | NEA's Involvement | Boundary[3] | GRI[4] Topic-specific Standards |
|---|---|---|---|---|
| **High Public Health Standards** | **Why topic is material**<br>Good control of vector-borne diseases, management of public cleanliness, environmental sanitation and hygiene, enforcement of smoking-prohibited places, and provision of government after-death facilities and services are necessary for maintaining high public health standards vital to Singapore residents' quality of life<br>**Where the impact occurs**<br>• Public • Community | • Enhance vector control and sustain low endemicity of vector-borne diseases through surveillance, prevention and control, outbreak management, enforcement, research and outreach<br>• Achieve low incidence of public health issues involving waterborne pathogens, and improve hygiene standards of high-risk premises and toilets<br>• Minimise exposure to second-hand tobacco smoke in public places<br>• Ensure sufficient capacity of after-death facilities, and explore provision of additional after-death options | External | Non-GRI |
| **Safe, Healthy and Conducive Living Environment** | **Why topic is material**<br>Monitoring, reducing and preventing environmental pollution, as well as incorporating environmental factors into land use planning, development and building control of new developments, are vital in ensuring a safe, healthy and conducive living environment<br>**Where the impact occurs**<br>• Public • Community | • Maintain clean air through effective monitoring capabilities and implementation of policies<br>• Maintain the quality of coastal and inland waters for the public's enjoyment<br>• Maintain clean land through sound management of hazardous substances<br>• Safeguard the public, radiation workers and the environment against the harmful effects of radiation<br>• Develop standards and mitigation measures to address construction noise | External | • Emissions<br>• Effluents<br>• Environmental Compliance |
| | | • Maintain clean air with low levels of pollutants at our incineration plants, comparable with the best cities in the world | Internal | |

[3] *Internal material issues refer to issues that are directly managed and controlled within NEA. External material issues refer to issues beyond NEA's direct control, but is managed by NEA to improve the external environment*
[4] *GRI Sustainability Reporting Standards are one of the most widely used standards for sustainability reporting. More information on the GRI Standards can be found on www.globalreporting.org/standards/*

| Material Issues | Impact on NEA and Stakeholders | NEA's Involvement | Boundary | GRI Topic-specific Standards |
|---|---|---|---|---|
| **Sustainable and Resource Efficient Singapore** | **Why topic is material**<br>Minimising waste generation and disposal, increasing energy efficiency, and water management are vital in conserving landfill space and resources<br>**Where the impact occurs**<br>• Public • Community | • Move towards a Zero Waste Nation and circular economy<br>• Strengthen corporate environmental responsibility and promote the 3Rs to conserve resources<br>• Promote energy efficiency and conservation to reduce GHG emissions and mitigate climate change | External | • Energy<br>• Emissions<br>• Waste |
| | | • Develop and manage solid waste management facilities<br>• Promote energy and water conservation as well as waste reduction within NEA | Internal | |
| **Timely and Reliable Weather and Climate Services** | **Why topic is material**<br>Meteorological services are vital in providing valuable weather and climate services to support informed decision-making and planning<br>**Where the impact occurs**<br>• Public • Community | • Deliver timely and reliable weather forecast and warning services customised to user needs and requirements<br>• Conduct research to advance understanding and prediction of the weather and climate of Singapore and the region<br>• Strengthen climate science capabilities to support climate adaptation planning | External | Non-GRI |
| **Vibrant Hawker Culture** | **Why topic is material**<br>Hawker centres are community dining rooms where people from all walks of life can enjoy affordable food in a clean and hygienic environment<br>**Where the impact occurs**<br>• Industry • Public • Community | • Ensure availability of affordable cooked food options in a clean and hygienic environment<br>• Sustain the hawker trade by supporting new entrants and existing hawkers with training and technology adoption<br>• Drive initiatives for environmentally sustainable and productive hawker centres<br>• Enhance vibrancy of hawker centres through place-making efforts, and promote gracious and considerate behaviour among patrons | External | Non-GRI |
| **Private Sector as Force Multipliers for Mission Success** | **Why topic is material**<br>Environmental Services (ES) provide essential cleaning, waste management and pest management services to keep Singapore clean and hygienic<br>**Where the impact occurs**<br>• Industry • NGOs • Partners • Public Sector Agencies | • Build strong capabilities and spearhead technology and innovation in Singapore's ES industry<br>• Improve land use efficiency and manpower productivity<br>• Support local companies to become global players, and promote local companies internationally<br>• Build a skilled and resilient workforce for a vibrant, sustainable and professional ES industry | External | Non-GRI |

| Material Issues | Impact on NEA and Stakeholders | NEA's Involvement | Boundary | GRI Topic-specific Standards |
|---|---|---|---|---|
| **Active Community as Stewards of the Environment** | **Why topic is material**<br>Collaboration with stakeholders and public support are critical to NEA's mission success<br>**Where the impact occurs**<br>• NGOs • Industry • Public • Community • Partners | • Build public support for NEA as a trusted environmental authority, working in collaboration with the public<br>• Co-create solutions with citizens, businesses and NGOs<br>• Grow, nurture and engage volunteers<br>• Inculcate values, shape positive social norms and practices in the community that promote the adoption of environmentally friendly habits as a way of life | External | Non-GRI |
| **Public Sector Taking the Lead in Environmental Sustainability** | **Why topic is material**<br>A Whole-of-Government (WOG) approach is needed to ensure a clean and sustainable Singapore<br>**Where the impact occurs**<br>• Public Sector Agencies | • Co-drive sustainability efforts across the public sector | External | Non-GRI |
| **International Partnerships as Strategic Levers** | **Why topic is material**<br>International engagement is vital in safeguarding and advancing Singapore's environmental interests<br>**Where the impact occurs**<br>• NGOs • Partners • Public Sector Agencies<br>• Public • Community | • Develop and implement strategies to advance Singapore's environmental interests vis-à-vis NEA's strategic partners<br>• Develop collaborations, initiatives and engagements with strategic partners, such as foreign governments, NGOs and international organisations, in technical, scientific and policy areas of benefit to NEA<br>• Implement Singapore's obligations under the relevant multilateral environmental agreements | External | Non-GRI |
| **Creation of Capacity** | **Why topic is material**<br>Operational capacity is vital for service delivery and enforcement efforts, especially since NEA is an agency with multiple public touchpoints and heavy operational workload<br>**Where the impact occurs**<br>• NGOs • Partners • Public Sector Agencies<br>• Public • Community | • Leverage technology to enhance operations and enforcement, as well as explore shared enforcement modes to improve efficiency<br>• Harness the use of data analytics and behavioural insight strategies to improve service delivery efforts<br>• Right-size regulations to improve efficiency of business processes | Internal | Non-GRI |

| Material Issues | Impact on NEA and Stakeholders | NEA's Involvement | Boundary | GRI Topic-specific Standards |
|---|---|---|---|---|
| **Strong Capabilities for Mission Success** | **Why topic is material**<br>Digitalisation and capability development are key pillars in meeting public sector transformation objectives<br>**Where the impact occurs**<br>• Employees | • Build scientific, technological, engineering and infrastructural capabilities to support domain areas and achieve strategic outcomes<br>• Improve process and productivity through planning and designing a unified, standardised and flexible information technology (IT) infrastructure and capability<br>• Share information and knowledge that enable value creation by NEA, partners and stakeholders | Internal | Non-GRI |
| **Strong Corporate Practices for a Sustainable Organisation** | **Why topic is material**<br>Financial management and manpower management are vital in enabling effective and efficient operations, governance and development. Good procurement practices are vital in enabling fair, transparent, effective and efficient operations and governance<br>**Where the impact occurs**<br>• Employees • Industry • NGOs | • Responsible utilisation of financial resources that aims for efficiency and budget marksmanship, while delivering effective results within operating resource constraints<br>• Proactive planning to anticipate and identify the shape and structure of NEA's future workforce to ensure that there is sufficient and sustainable capacity and capabilities to fulfil NEA's objectives now and into the future<br>• Comprehensive system of internal controls based on sound principles to provide guidance, direction and oversight in the running of NEA, coupled with a proactive and integrated approach to manage uncertainties that may negatively affect NEA's objectives | Internal | • Economic Performance<br>• Procurement Practices<br>• Employment<br>• Occupational Health and Safety |
| **Conducive Culture for Collaboration, Learning and Innovation** | **Why topic is material**<br>A strong organisational culture is vital for a high performance and future-ready organisation<br>**Where the impact occurs**<br>• Employees | • Strong emphasis on training and development to ensure that employees are equipped with the right competencies to contribute effectively in their career with NEA<br>• Steer NEA towards becoming a learning organisation, by encouraging self-directed mobile learning via the LEARN app, recognising learning contributions through LEARNing and Trainer Awards, and promoting the sharing of NEA knowledge assets via the EMPOWER portal on the intranet<br>• Reach out to all employees to address their concerns, health and well-being so as to build an engaged workforce<br>• Build shared values, beliefs and behaviours that enable our staff to anticipate and adapt to change and disruptions | Internal | Training and Education |

# RISKS AND OPPORTUNITIES

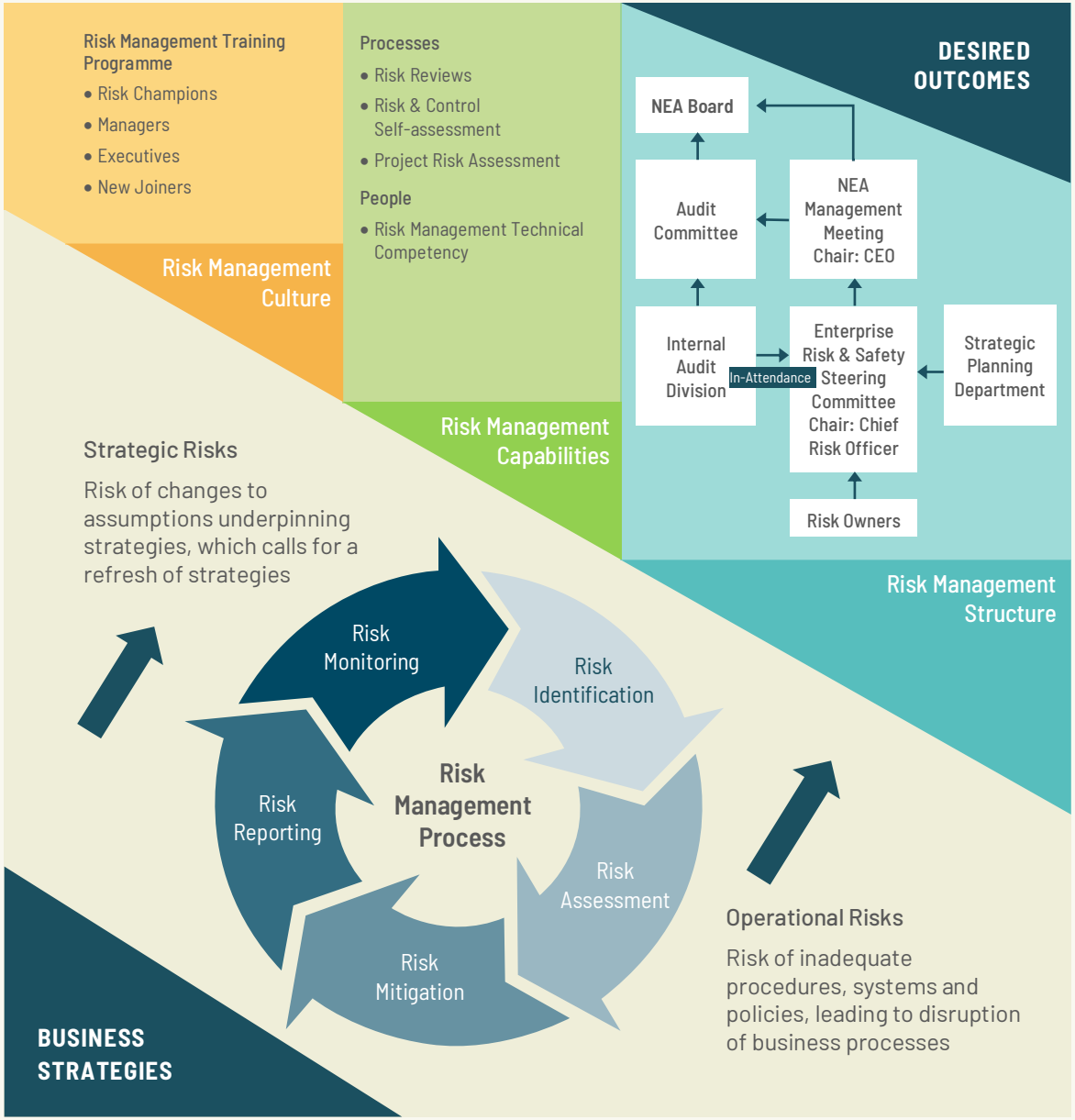
## ENTERPRISE RISK MANAGEMENT FRAMEWORK

Recognising the importance of effective risk management in enhancing corporate governance and strategic planning, NEA has established an Enterprise Risk Management (ERM) framework to manage risks and opportunities in a structured, integrated and effective manner.

The ERM framework enables NEA to proactively identify, assess, prioritise, treat and monitor our key strategic and operational risks on an ongoing basis. It enhances:

- Risk Awareness, by providing an integrated view of the combined impact of different risks on the organisation, thereby eliminating silo thinking in risk management
- Risk Ownership, by providing a formalised structure to assign accountability for key risks, mitigating measures and treatment plans
- Proactive Risk Management, by providing a structured and robust management process to minimise upstream risks, based on both top-down and bottom-up risk reviews

NEA takes a precautionary approach to managing our risks and closely monitors them, as well as identifies new risks and opportunities. This is to ensure that NEA's corporate sustainability efforts address all developing trends and risks.



In planning ERM activities, NEA is guided by an ERM framework that features three thrusts:

### Robust Risk Management Governance Structure

The Enterprise Risk and Safety Steering Committee (ERSSC) is responsible for driving overall ERM efforts in NEA, and comprises representatives across NEA. It is chaired by the Deputy CEO (Planning, Corporate and Technology) in his capacity as NEA's Chief Risk Officer.

The ERSSC meets four times a year, and reviews key strategic and operational risks faced by NEA. The ERSSC also reviews the effectiveness of these risk treatment plans to ensure key risks are adequately addressed, as well as studies trends and emerging issues that may impact the risks confronting NEA.

The ERSSC updates the CEO, Audit Committee and Board on strategic risks which NEA faces. Risk Work Groups are formed to comprehensively cover our work scope, aligned with NEA's structure and strategy map. Risk Owners are assigned the task of identifying, assessing, and monitoring key strategic and operational risks, and treatment plans to mitigate those risks. As part of the risk management process, environmental scans are conducted on material issues that may impact the achievement of strategic objectives in the strategy map. Meanwhile, Process Owners provide risk content such as root causes, consequences and controls, as well as implement and monitor the risk treatment plans.

### Proactive Risk Management Culture

To strengthen our risk culture, NEA conducts risk management training sessions and customised briefings each year. For example, risk management training is conducted as part of middle management's Leaders in Environmental Action Planning course and the NEA Executive Training course for all executive-grade staff, to create greater awareness and understanding of risk management, and how staff can play a part in instilling risk management practices in their areas of work.

Risk Champions, who are nominated by their respective division or department heads to facilitate risk-related initiatives within their division or department, are given due recognition for their contributions in their staff appraisal.

### Strong Risk Management Capabilities and Tools

NEA constantly strengthens our risk management capabilities, which comprise processes and people. Processes such as risk reviews, project risk assessment, and risk and control self-assessment are in place to ensure risk and control issues are proactively managed. Structured training programmes train our people in risk management tools, methodology and templates.

## MATERIALITY ASSESSMENT

The materiality assessment, which is part of NEA's ERM framework, enables us to identify and assess our environment, social and governance risks, challenges and opportunities.

| Material Issues | Risks and Challenges | Opportunities |
|---|---|---|
| **High Public Health Standards** | • Vector-borne disease outbreaks | • Use of data to understand epidemiology of diseases for temporal and spatial risk stratification<br>• Leverage technology to enhance our operations and build capability in related areas of research |
| **Safe, Healthy and Conducive Living Environment** | • Deterioration of ambient air quality<br>• Deterioration of coastal and inland water quality | • Enhance capabilities in air and water quality monitoring and pollution source identification with technology |
| **Sustainable and Resource Efficient Singapore** | • Resource scarcity<br>• Insufficient waste management capacity<br>• Climate change | • Drive infrastructure enhancement, innovation and behavioural changes to achieve environmental sustainability<br>• Gain public trust for our national policy by walking the talk |
| **Timely and Reliable Weather and Climate Services** | • Severe weather conditions | • Leverage technology to boost our capabilities in weather monitoring and forecasting |

| Material Issues | Risks and Challenges | Opportunities |
|---|---|---|
| **Vibrant Hawker Culture** | • Inability to meet the need for affordable cooked food options in a clean and hygienic environment | • Build new hawker centres and rejuvenate existing ones<br>• Offer a variety of pathways for aspiring hawkers |
| **Private Sector as Force Multipliers for Mission Success** | • Inability to meet increasing demand for Environmental Services (ES) | • Transform the ES industry through the Environmental Services Industry Transformation Map, by helping the industry adopt technology to improve standards and service delivery, build a skilled productive workforce, as well as grow and export their businesses |
| **Active Community as Stewards of the Environment** | • Lack of environmental ownership | • Work more closely with NEA's partners to achieve environmental ownership |
| **Public Sector Taking the Lead in Environmental Sustainability** | • Resource scarcity<br>• Insufficient waste management capacity<br>• Climate change | • Drive innovation and behavioural changes to achieve environmental sustainability<br>• Gain public trust for our national policy by walking the talk |
| **International Partnerships as Strategic Levers** | • Inability to fulfil NEA's obligations set out in environmental agreements | • Increase in demand for clean energy, low-emissions transport and sustainable urban solutions |
| **Creation of Capacity** | • Compromised data due to malicious activities against NEA's IT systems and infrastructure<br>• To continue to be relevant in the face of evolving needs and challenges | • Deepen technical capabilities in key areas to improve our services<br>• Simplify application and compliance processes of NEA's regulations where feasible |

| Material Issues | Risks and Challenges | Opportunities |
|---|---|---|
| **Strong Capabilities for Mission Success** | • Inability to meet NEA's manpower requirements | • Uncover new areas for collaboration and deployment of systems and services<br>• Improve productivity and forge a future-ready NEA |
| **Strong Corporate Practices for a Sustainable Organisation** | • Any unmet rising public expectations<br>• Weak controls resulting in fraudulent practices<br>• Challenges in recruitment and retention of talent<br>• Instances of non-compliance with Workplace Safety and Health regulations or procedures | • Spur accountability and resource optimisation<br>• Gain public's confidence in how NEA manages our finances as a government-funded organisation<br>• Encourage strong corporate governance and risk management practices<br>• Gain public confidence that NEA uses our resources responsibly<br>• Enable a high-performance NEA by driving organisational productivity and growth, and strengthen our brand as an employer of choice<br>• Develop a stronger sense of identity and purpose among staff<br>• Develop a more engaged and socially responsible workforce that gives back to the community |
| **Conducive Culture for Collaboration, Learning and Innovation** | • Teams work in silos, and data sharing is fragmented | • Enable a high-performance NEA by driving organisational productivity and growth, and strengthen our brand positioning as an employer of choice |

# Ensuring a Clean and Sustainable Environment for Singapore

## HIGH STANDARDS OF PUBLIC HEALTH

**As the national authority on environmental public health standards in Singapore, NEA undertakes a critical role to maintain these standards through research, ground surveillance, enforcement, licensing and outreach.**

### MONITORING INFECTIOUS DISEASE THREATS

Singapore was one of the early adopters of tracking COVID-19 cases with wastewater surveillance, to detect and stem the growth of potential clusters. This was done by screening wastewater samples for the SARS-CoV-2 virus, which could be shed by infected individuals through their stools or sputum. From a modest eight sites in May 2020, NEA progressively expanded surveillance coverage to more than 500 sites across Singapore in 2022. We also scaled up our laboratory testing capacity through partnerships with local academia and private sector laboratories, and further enhanced our in-house laboratory capabilities with an automated liquid handling system co-developed with the Agency for Science, Technology and Research. As Singapore transits to COVID-19 endemicity, NEA's focus on wastewater testing has shifted from early case detection to providing situational awareness on whether infections among the population are increasing or decreasing, independent of prevailing clinical test protocols and population healthcare-seeking behaviour. Wastewater surveillance provides a complementary, objective and independent indicator of the trajectory of COVID-19 infections. Beyond COVID-19, we are also looking to extend the use of the wastewater surveillance system to study other infectious disease threats.

### KEEPING DENGUE AT BAY

Singapore reported a significant decrease in dengue cases in 2021 with 5,258 cases – almost seven times lower than the 35,266 cases reported in 2020. This was likely due to the temporary cross-protective immunity arising from the large dengue outbreak from 2019 to 2020.

In 2021, NEA conducted 631,000 inspections for mosquito breeding, of which 6,300 were at construction sites. A total of 11,000 enforcement actions were taken against premises owners and occupiers for mosquito breeding, which included 690 enforcement actions and 43 Stop Work Orders issued to construction sites, and 64 contractors charged in court for repeat offences.

NEA launched the annual National Dengue Prevention Campaign in April 2021, ahead of the traditional peak dengue season, to rally community efforts in the fight against dengue. We introduced a new acronym 'B-L-O-C-K' to help the public remember the Mozzie Wipeout steps and common mosquito breeding areas, and refreshed the dengue cluster alert banners for greater public visibility. We also embarked on a nationwide publicity campaign on mass media channels and digital platforms to highlight how dengue could lead to severe health consequences.

Our volunteers and community partners supported with outreach activities to educate residents about breaking the dengue transmission chain. As most residents were still working and studying from home, dengue prevention reminders were shared on social media and chat platforms to raise greater awareness.

However, February 2022 saw a resurgence of dengue cases, likely attributed to the waning of the cross-protective immunity and coupled with the higher mosquito population and circulation of the less-common DENV-3 serotype. Responding to early signs of an outbreak, we activated the Inter-agency Dengue Task Force for intensive source reduction efforts in anticipation of a rise in cases. The National Dengue Prevention Campaign was once again brought forward to March 2022, where the new purple community alert banners were launched as part of the colour-coded Dengue Community Alert System to inform residents of the dengue situation in their estate. The purple alert banners were placed in non-dengue cluster areas with a persistently high *Aedes aegypti* mosquito population, serving as a visual reminder for proactive mosquito prevention measures to minimise the risk of dengue cluster formation.

NEA engaged public agencies, Town Councils, and pest control and construction industries, to implement timely and regular intervention measures in their premises. These outreach efforts were amplified through extensive publicity on online and traditional media for community awareness.

## MITIGATING AEROSOL TRANSMISSION RISKS

Improving ventilation and indoor air quality in built environments is critical to mitigate the risk of transmission of diseases like COVID-19, which can be spread through virus aerosols in the air. Together with the Building and Construction Authority (BCA) and Ministry of Health, we worked with various government agencies to conduct field investigations to understand aerosol transmission risk in the built environment and recommend mitigation measures. The premises covered included hotels, preschools, Changi Airport, nursing homes, polyclinics, hawker centres and markets. We also worked with agencies to take carbon dioxide ($CO_2$) readings at over 2,700 sampling locations, to assess if indoor spaces were sufficiently ventilated. Findings and recommendations were communicated to agencies and premises operators, including through customised webinars that reached an estimated 3,000 premises operators across various sectors. In addition, we developed advisories on $CO_2$ monitoring and improving ventilation and indoor air quality amid the pandemic, as well as guidelines and whitelists for disinfectant and air cleaning products and technologies, to guide consumer decision-making.

## SAFEGUARDING AGAINST WATERBORNE DISEASES

Following the amendments to the Environmental Public Health Act on 31 August 2021, premises owners and occupiers with Aerosol-Generating Systems, such as cooling towers, are required to register their cooling towers. In addition, higher-risk aquatic facilities (AF) such as water playgrounds, interactive water fountains and multi-use spa pools are required to be licensed. Prior to this, only swimming pools were licensed.

Premises owners and occupiers of cooling towers and AF are required to carry out water quality tests periodically to ensure that the water is properly disinfected to prevent transmission of waterborne infectious diseases. Test results are to be submitted to NEA for regulatory compliance, serving as early alerts to disease outbreak.

To keep the industry informed of these changes and regulatory requirements, we engaged 635 industry stakeholders in August 2021. The updated AF design requirements and existing requirements for cooling towers were incorporated into the Code of Practice on Environmental Health, which was published on 1 September 2021.

In tandem, the NEA ePortal was enhanced to enable registration and submission of test results for cooling towers, while the GoBusiness Licensing portal was enhanced to enable the licensing and submission of test results for AF.

## RAISING ENVIRONMENTAL SANITATION STANDARDS

Since 30 July 2021, the Environmental Sanitation regime has been progressively implemented in high-risk premises with vulnerable occupants or high footfall to ensure a safe environment. These include eldercare, youth and social service facilities, preschools, Ministry of Education schools, food centres and markets, and coffeeshops. The regime builds on existing hygiene and sanitation measures, by providing clear guidance on the minimum cleaning and disinfection frequencies and scope to ensure premises are properly cleaned and disinfected at all times.

The Code of Practice for Environmental Control Coordinators (ECC) was first published on NEA's website in July 2021, with the second iteration published in December 2021. It details the roles and responsibilities of ECCs and Premises Managers, and guides ECCs in developing and executing an environmental sanitation programme for specified premises.

From 1 April 2022, the regime has been progressively extended to more premises which are publicly accessible, experience high footfall, or are congregation areas.

## STEPPING UP CLEANING AND DISINFECTION

Proper cleaning and disinfection is necessary for public safety and to prevent the spread of diseases, particularly in the aftermath of an infectious disease outbreak. When the COVID-19 pandemic hit, NEA's High Readiness Core team stepped up to oversee and provide on-site guidance for disinfection operations, covering more than 12,000 premises exposed to COVID-19 cases. For residential premises, we provided guidelines and advice on how to clean and disinfect homes effectively. We also handed out cleaning kits to people who were under quarantine and unable to leave their houses. For non-healthcare premises, we engaged premises owners to provide on-site guidance on the disinfection measures. NEA also collaborated with the Ministry of Manpower (MOM), Enterprise Singapore (ESG) and Singapore Tourism Board to reach out to more stakeholders, such as dormitory operators, hotels, building managements and Town Councils, to carry out disinfection of their premises and common areas visited by confirmed cases.

### IMPROVING CLEANLINESS OF PUBLIC TOILETS

The Toilet Improvement Programme aims to improve public toilet cleanliness and encourage good personal hygiene among users. Under the programme, NEA co-funds toilet improvement works in coffeeshops and hawker centres to improve user experience and enhance public hygiene.

As part of the Coffeeshop Toilet Improvement Programme, NEA helps coffeeshop operators and owners defray the cost of toilet improvement works, which include the adoption of new technologies that facilitate easier cleaning and maintenance. Those eligible can receive co-funding support for renovation, deep cleaning, and the training of cleaners.

Similarly, the Hawker Centre Toilet Improvement Programme provides support to Town Councils to improve public toilets in the hawker centres under their management. The programme focuses on improving toilet design and features, and adopting toilet technologies to facilitate easier cleaning and maintenance of toilets.

To ensure high hygiene and cleanliness standards of public toilets, NEA will continue to conduct inspections and enforcement against premises owners and operators for lapses.

## GOING FORWARD

In the short term, NEA's focus is to raise cleanliness and environmental hygiene standards for public health, which is vital in the ongoing COVID-19 pandemic. Beyond this, we are also building a national wastewater epidemiology and testing ecosystem, to serve as an additional surveillance indicator to assess the spread of disease in a community. We will continue to tackle high-rise littering and strengthen our partnership with stakeholders to keep Singapore clean. We will also look into improving indoor air quality.

Our medium-term priority is to develop new capabilities to counter the dengue threat and enhance vector control operations. We endeavour to scale up key transformative initiatives such as Project *Wolbachia*, which has seen promising results in the fight against dengue.

Our priority on the after-death front is to ensure the adequate provision of after-death facilities, and enhance service standards of the funeral industry.

NEA's vision is a clean and healthy Singapore with high public health standards, vital for a good quality of life for residents.

### SUPPRESSING DENGUE WITH PROJECT *WOLBACHIA*

Novel vector control tools, such as Project *Wolbachia* – Singapore, will continue to complement conventional dengue control measures. In October 2021, small-scale test releases of non-biting male *Wolbachia-Aedes* mosquitoes commenced in the Marine Parade landed residential estate. This was to understand the behaviour of the released male *Wolbachia-Aedes* mosquitoes in such an environment, before a larger-scale deployment in April 2022 to cover five square kilometres within the estate. NEA also carried out releases at a few construction sites, as part of an overall strategy to achieve better suppression by covering different terrains and landscapes.

Since April 2022, releases of male *Wolbachia-Aedes* mosquitoes at Tampines and Yishun have been expanded to cover the entire towns. High-risk areas in Choa Chu Kang and Bukit Batok also continued to see targeted releases of male *Wolbachia-Aedes* mosquitoes. The results have been promising, with up to 98 per cent reduction in the *Aedes aegypti* mosquito population and up to 88 per cent reduction in dengue cases observed in Tampines and Yishun, after more than a year of releases.

### EXPANDING OUR MEMORIAL FACILITIES

The Inland Ash Scattering Garden at Choa Chu Kang Cemetery Complex was launched in May 2021 to provide Singaporeans with an additional option of managing cremated human remains in Singapore. Named the Garden of Peace @ CCK, NEA received about 900 applications for the scattering of ashes in 2021.

We also plan to introduce inland ash scattering services at the Garden of Serenity in the new Mandai Crematorium Complex. This will be completed together with the new complex, and is expected to be operational in 2024. More cremators and building fittings will be progressively installed in tandem with demand.

# SAFE, HEALTHY AND CONDUCIVE LIVING ENVIRONMENT

**NEA enforces strict regulations in line with international standards to limit emissions, and implements various initiatives to monitor, prevent and mitigate environmental pollution and radiation. NEA plays a pivotal role to sustain clean air, good water quality and quieter surroundings for all Singaporeans.**

## COMBATING AIR POLLUTION

To ensure that the incineration processes at our waste-to-energy plants and crematoria do not pollute the environment, the facilities must comply with Singapore's Environmental Protection and Management (Air Impurities) Regulations[5] to manage their air emissions.

The design features incorporated into the Tuas Incineration Plant and Tuas South Incineration Plant ensure that flue gas emissions, which are cleaned prior to release through the chimneys, meet regulatory limits. Efficient process control systems are in place to regulate the rate of air distribution and refuse feed. These ensure proper combustion in the furnace, leading to greater fuel efficiency and lower emissions.

In addition, NEA installed a telemetric monitoring system that continuously monitors air emissions from these plants. NEA also monitors air emissions from the Mandai Crematorium, to ensure that emissions are kept within limits[6].

In FY2021, the air emissions from the three facilities were within regulatory limits, and the facilities did not have any black smoke or air pollution incidents. NEA has implemented an effective environmental management system, with processes and practices to boost NEA's operational efficiency and minimise environmental impact. We review our performance periodically to ensure alignment with our sustainability goals.

**Material Air Emissions**

| NEA's Facilities | Compounds ($mg/Nm^3$) | FY19 | FY20 | FY21 | Emission Limit[7] |
|---|---|---|---|---|---|
| Tuas Incineration Plant | Dust | 13.8 | 16.2 | 15.5 | < 50 |
| | $SO_2$ | 84.2 | 168.7 | 97.0 | < 1,700 |
| | $NO_x$ | 346.7 | 352.5 | 305.4 | < 400 |
| Tuas South Incineration Plant | Dust | 7.0 | 7.8 | 7.5 | < 50 |
| | $SO_2$ | 91.4 | 83.3 | 92.1 | < 1,700 |
| | $NO_x$ | 166.7 | 141.2 | 179.1 | < 400 |
| Mandai Crematorium | Dust | 15.0 | 6.8 | 16.4 | < 50 |
| | $SO_2$ | < 0.1 | 7.7 | 0.2 | < 1,700 |
| | $NO_x$ | 362.8 | 125.7 | 86.0 | < 400 |
| | HCl | < 0.1 | 0.5 | 9.8 | < 200 |
| | CO | 4.3 | < 0.1 | 5.0 | < 250 |

## ENHANCING SPATIAL RESOLUTION OF AIR QUALITY MONITORING

NEA continuously monitors air quality across Singapore through a network of ambient air monitoring systems, providing hourly updates on air quality information on the NEA website and myENV App.

In 2020, we commenced a two-year pilot to enhance air quality monitoring and air pollution control efforts, particularly within localised areas such as residential estates. Using low-cost and low-footprint sensors, we can obtain air quality data at higher spatial and temporal resolution to enhance air quality monitoring. The sensors provide useful insights into the levels and dispersion of air pollutants at different times of the day or during monsoon seasons, at different heights and areas within an estate. This data is then displayed in a heat map for easy visualisation. Such localised monitoring will enable better support for our air quality monitoring and air pollution control efforts.

[5] *Air emission limits extracted from: sso.agc.gov.sg/SL/EPMA1999-RG8*
[6] *NEA conducts annual source-emission testing for the Mandai Crematorium, to assess compliance with the emission limits indicated in the Environmental Protection and Management (Air Impurities) Regulations*
[7] *This set of emission limits reflects the updated limits from Singapore's Environmental Protection and Management (Air Impurities) Regulations, which has been effective since 1 July 2015. Existing plants have until 1 July 2023 to comply with the revised standards for particulate matter, nitrogen oxides, carbon monoxide and sulphur dioxide*

**Air Quality**

| Key Performance Indicators (KPI) | 2020 Results | 2021 Results | Targets[8] |
|---|---|---|---|
| $PM_{2.5}$ (annual mean) | 11.0 µg/m$^3$ | 12.0 µg/m$^3$ | 12.0 µg/m$^3$ |
| $PM_{2.5}$ (24-hour mean, 99$^{th}$ percentile) | 24.0 µg/m$^3$ | 28.0 µg/m$^3$ | 37.5 µg/m$^3$ |
| $PM_{10}$ (annual mean) | 25.0 µg/m$^3$ | 28.0 µg/m$^3$ | 20.0 µg/m$^3$ |
| $PM_{10}$ (24-hour mean, 99$^{th}$ percentile) | 43.0 µg/m$^3$ | 51.0 µg/m$^3$ | 50.0 µg/m$^3$ |
| $SO_2$ (24-hour mean, max) | 30.0 µg/m$^3$ | 89.0 µg/m$^3$ | 50.0 µg/m$^3$ |
| $O_3$ (8-hour mean, max) | 145.0 µg/m$^3$ | 176.0 µg/m$^3$ | 100.0 µg/m$^3$ |
| $NO_2$ (annual mean) | 20.0 µg/m$^3$ | 25.0 µg/m$^3$ | 40.0 µg/m$^3$ |
| $NO_2$ (1-hour mean, max) | 118.0 µg/m$^3$ | 123.0 µg/m$^3$ | 200.0 µg/m$^3$ |
| CO (8-hour mean, max) | 1.2 mg/m$^3$ | 1.2 mg/m$^3$ | 10.0 mg/m$^3$ |
| CO (1-hour mean, max) | 1.6 mg/m$^3$ | 1.3 mg/m$^3$ | 30.0 mg/m$^3$ |

## MOVING TOWARDS CLEANER COMMERCIAL VEHICLES

In April 2021, the Commercial Vehicle Emissions Scheme (CVES) was introduced to encourage the adoption of cleaner new light commercial vehicles with lower emissions. Incentives are granted for cleaner models, while a surcharge is imposed on more pollutive models. The CVES works in tandem with the Early Turnover Scheme (ETS) that was introduced in April 2013, to promote the replacement of older, more pollutive diesel commercial vehicles and buses. The ETS provides a discount off the Prevailing Quota Premium when owners of older diesel commercial vehicles switch to cleaner new vehicles. Since the implementation of the CVES and ETS, around 80 per cent of light commercial vehicles registered in 2021 were of cleaner models, and over 50,000 commercial vehicles were replaced, as of March 2022.

## MONITORING WATER DISCHARGE

The quality of effluent discharge from NEA's two waste-to-energy plants comply with the allowable limits under the Sewerage and Drainage Act (Chapter 294, Sections 72 and 74) and Sewerage and Drainage (Trade Effluent) Regulations by PUB, Singapore's National Water Agency (PUB).

**Water Discharge**

| Amount of wastewater discharged (L) | FY19 | FY20 | FY21 |
|---|---|---|---|
| Tuas Incineration Plant | 21,058,000 | 18,829,000 | 19,997,000 |
| Tuas South Incineration Plant | 56,340,000 | 48,341,000 | 41,656,000 |

## ENFORCING CONSTRUCTION NOISE CONTROLS

In September 2021, the Environmental Protection and Management (Amendment) Bill was passed in Parliament to strengthen enforcement of the no-work rule on Sundays and public holidays. The no-work rule applies to worksites located within 150m from any hospital, home for the aged sick or residential building.

With effect from 1 October 2022, errant contractors that violate the no-work rule are required to install electronic video surveillance systems. The use of electronic video surveillance technology will enhance NEA's operational effectiveness by enabling our officers to monitor construction activities remotely without being physically present at the construction site, and deter future violations of the no-work rule.

## ANALYSING AIR AND WATER QUALITY TRENDS

NEA published our inaugural State of the Environment: Air and Water Quality Report[9] in September 2021, to present the trends and analyses of the air and water quality in Singapore, and showcase NEA's environmental monitoring capabilities. The publication addressed the impact of the COVID-19 circuit breaker on urban air quality, and the emerging contaminants in the tropical aquatic environment. It also highlighted the outcomes and applications of NEA's research collaboration with the NUS Environmental Research Institute, including the identification of the sources of biomass burning and $PM_{2.5}$ in Southeast Asia.

[8] *2020 targets are currently being revised and a new set of targets will be ready by 2023*
[9] *Read the State of the Environment: Air and Water Quality Report at go.gov.sg/nea-soe-report*

## GOING FORWARD

As a steward of the environment, NEA's short-term focus is to implement initiatives to monitor, prevent and mitigate air, land, water and noise pollution to sustain a quality living environment. Our medium-term priority is to heighten our emergency preparedness capabilities to safeguard against the harmful effects of radiation. NEA's vision is a safe and healthy Singapore that is vital for good quality of life for residents.

### ENCOURAGING CLEANER PASSENGER CARS AND TAXIS

The Vehicular Emissions Scheme (VES) encourages the purchase of cleaner passenger cars and taxis with lower emissions, by providing rebates for cleaner models and imposing surcharges on more pollutive models. The VES was enhanced with increased rebates from 1 January 2021 and increased surcharges from 1 July 2021. Since then, we have observed an increase in the proportion of cleaner cars registered under the A1 and A2 rebate bands, from 39 per cent in the first quarter of 2021 to 50 per cent in the last quarter of 2021. The enhanced VES will be extended till 31 December 2025, with tightened pollutant thresholds to be introduced from 1 January 2024. These enhancements are in line with the WOG push towards a cleaner energy vehicle fleet in Singapore over the next decade, under the Singapore Green Plan 2030.

### ALIGNING WITH INTERNATIONAL NOISE STANDARDS

Starting from 1 April 2023, NEA will exclusively adopt the latest UN noise standards for new vehicle registrations. The UN standards, which are generally more stringent than Singapore's current standards, involve the use of a test procedure that better reflects actual driving conditions and non-exhaust noise. The internationally-recognised UN standards have been adopted by the European Union and Japan since 2016.

## SUSTAINABLE AND RESOURCE EFFICIENT SINGAPORE

**With Singapore's limited land and dense urban population, NEA prioritises the effective and targeted management of our respective waste streams as we progress towards a Zero Waste vision and circular economy, contributing to the Singapore Green Plan 2030.**

### SETTING TARGETS TO REDUCE AND RECYCLE

Under Singapore's Zero Waste Masterplan, we aim to increase the national overall recycling rate to 70 per cent and reduce waste-to-landfill per capita per day by 30 per cent by 2030. This goal will be frontloaded to achieve a 20 per cent reduction in waste-to-landfill per capita per day by 2026 under the Singapore Green Plan 2030. Targets are set with plans of extending the lifespan of Singapore's only landfill beyond 2035.

We also have 2030 targets to increase non-domestic recycling rate to 80 per cent and domestic recycling rate to 30 per cent.

**Waste and Recycling Targets**

| KPI | 2019 Results | 2020 Results[10] | 2021 Results[11] | Targets |
|---|---|---|---|---|
| Overall recycling rate | 60% | 52% | 55% | 70% by 2030 |
| Domestic recycling rate | 20% | 13% | 13% | 30% by 2030 |
| Non-domestic recycling rate | 70% | 68% | 70% | 80% by 2030 |
| Domestic sector – waste disposed of per capita | 0.8 kg a day/person | 0.8 kg a day/person | 0.8 kg a day/person | 20% by 2026 |
| Non-domestic sector – waste disposed of per $bn GDP | 8.9 tonnes a day/GDP ($bn) | 8.2 tonnes a day/GDP ($bn) | 8.7 tonnes a day/GDP ($bn) | |

[10] *2020 waste and recycling statistics were impacted by COVID-19, as overall waste generated and overall waste disposed of fell by 19 per cent and five per cent respectively as compared to 2019. For more details, visit www.nea.gov.sg/media/news/news/index/reduction-in-overall-waste-generation-in-2020-with-less-waste-sent-to-semakau-landfill*

[11] *The resumption of economic activities and normalisation of business and social activities over 2021 led to a 10 per cent and 26 per cent increase in the amounts of waste disposed of and waste recycled respectively, as compared to 2020. In comparison to the pre-COVID-19 pandemic period in 2019, 2021 saw five per cent more waste disposed of and 11 per cent less waste recycled. For more details, please visit www.nea.gov.sg/our-services/waste-management/waste-statistics-and-overall-recycling*

## HARNESSING WASTE PROCESS SYNERGIES

As part of NEA's long-term plan to meet Singapore's solid waste management needs, we collaborated with PUB in 2013 for the planning and development of the Integrated Waste Management Facility (IWMF). We embarked on a joint preliminary engineering design study to assess the potential of harnessing process synergies by co-locating a solid waste treatment facility with a used water reclamation facility. This led to the co-location of two mega facilities – the IWMF and Tuas Water Reclamation Plant, now collectively known as Tuas Nexus – which will be able to reap co-location synergies across the water-energy-waste nexus to optimise energy and resource recovery, and land use.

Developed in phases, we have since made several important strides in this greenfield project. Under the first phase, NEA commenced the construction of the Waste-to-Energy Facility (2,900 tonnes/day) and Materials Recovery Facility (250 tonnes/day) in May 2020, and the Sludge Incineration Facility (800 tonnes/day) and Food Waste Treatment Facility (400 tonnes/day) in August 2021. Phase 1 is expected to be completed in 2025 to 2026, with the development of Phase 2, which comprises another waste-to-energy Facility (2,900 tonnes/day), to follow thereafter.

In August 2021, NEA also announced the establishment of our $3 billion Multicurrency Medium Term Note (MTN) Programme and Green Bond Framework. Proceeds from the issuance of Notes under the MTN Programme will be used to finance sustainable infrastructure development projects including the IWMF.

## CLOSING THE WASTE AND RESOURCE LOOPS

The Closing the Waste Loop (CTWL) Funding Initiative (FI) is a research and development (R&D) programme announced in December 2017, along with the Environmental Services Industry Transformation Map. NEA has been administering the $45 million CTWL FI to develop innovative solutions in resource recovery, conservation of landfill space, and sustainability of our waste management system. As of 31 March 2022, we have committed $37 million to fund 15 R&D projects on the treatment of and resource recovery from priority waste streams – electrical and electronic waste (e-waste), food waste and plastic waste, and diversion of residues such as incineration bottom ash and non-incinerable waste from Semakau Landfill. Several projects have attracted industry interest, leading to 10 industry project collaborations and $2.1 million industry spending pledged to date.

One of these innovative solutions was developed by the Nanyang Technological University, Singapore-CEA Alliance for Research in Circular Economy (SCARCE), which employs green chemical processes that avoided the use of hazardous substances to recover precious metals from e-waste. Using fruit peel, SCARCE has successfully recovered 90 per cent weight of metals, such as nickel, manganese, cobalt and lithium, from spent lithium-ion batteries.

In February 2022, NEA secured $80 million for the Closing the Resource Loop FI, which will build upon the R&D efforts under the earlier Waste-to-Energy FIs and the CTWL FI. This will support waste-related initiatives under the Singapore Green Plan 2030, through research and innovations.

## TAKING E-WASTE RECYCLING TO THE NEXT LEVEL

On 1 July 2021, NEA implemented the Extended Producer Responsibility scheme for e-waste management with multiple collection avenues islandwide. The move encourages the public to recycle their e-waste, such as IT equipment, large appliances, light bulbs and tubes, and batteries. The five-year e-waste collection contract was awarded to ALBA E-Waste Smart Recycling Pte Ltd. As the appointed operator of the Producer Responsibility Scheme, ALBA will send all e-waste collected to licensed e-waste recyclers.

## EXPLORING CHEMICAL RECYCLING SOLUTIONS

NEA is exploring chemical recycling solutions to close the plastic waste loop in Singapore. Chemical recycling is able to process plastics that cannot be mechanically recycled, such as contaminated single-use plastics, and convert these into higher-value products such as pyrolysis oil, that can be used as feedstock to manufacture new plastics and chemicals.

To gain a better understanding of the technical and commercial aspects of a chemical recycling value chain in Singapore, a joint NEA-Shell study explored the feasibility of chemically recycling plastic, and concluded that chemical recycling of plastic waste was technically viable with the appropriate plastic feedstock. There would also be carbon savings from the diversion of plastic waste from incineration. Concurrently, we embarked on a feasibility study for a Plastic Recovery Facility (PRF). As there are currently no local facilities to recover contaminated plastics from domestic waste, the PRF will fill this gap by taking in domestic waste collected from households and trade premises, to sort and recover various plastic polymer types such as Polyethylene and

Polypropylene for further treatment in chemical recycling plants. Completed in end-2021, the study provided an understanding of the design, suitable waste management technologies and equipment, as well as site requirements and scalability potential for a PRF. NEA will further assess the findings from both studies before proceeding with the next stage of development.

### IMPROVING INDUSTRY ENERGY PERFORMANCE

On 22 December 2021, the energy management system (EnMS) requirements took effect as an enhancement of the Energy Conservation Act to support Singapore's climate pledge under the Paris Agreement. Energy-intensive companies in the industrial sector, accounting for more than 60 per cent of Singapore's GHG emissions, are required to implement an EnMS as part of corporate energy management practices, to achieve continual improvement in energy performance. The EnMS is based on the Plan-Do-Check-Act cycle and comprises key elements of ISO 50001 - Energy Management System standard. Companies operating the most energy-intensive facilities ($\geqslant$500TJ/year) have submitted their first EnMS reports by 31 December 2021, while the remaining companies operating the next tier of energy-intensive facilities will submit their EnMS reports by 31 December 2022.

## GOING FORWARD

In the short-term, NEA's focus is to implement regulatory measures for our priority waste streams, develop our chemical recycling capabilities, and increase recycling rates. We also seek to reduce GHG emissions to meet our obligations under the Paris Agreement, and enhance energy efficiency efforts in households, industries and the private sector. Internally, we will drive initiatives within the organisation to align to GreenGov.SG.

Our medium-term priority is to innovate and develop the next generation of waste management infrastructure and capabilities, as we work towards becoming a Zero Waste Nation. NEA will also look into circular economy initiatives that seek to extend Semakau Landfill's lifespan, such as exploring the use of landfilled mixed material as reclamation fill, studying the development of a commercial NEWSand, and developing solutions for non-incinerable waste.

### MITIGATING THE IMPACT OF HYDROFLUOROCARBONS

In September 2021, the Environmental Protection and Management (Amendment) Bill was passed in Parliament to mandate measures to mitigate the GHG impact of hydrofluorocarbons (HFCs), which are used as refrigerants in Refrigeration and Air-conditioning (RAC) equipment.

The regulatory measures, which come into effect from October 2022, will restrict the supply of RAC equipment that use high Global Warming Potential refrigerants, and require at least one certified technician to supervise or carry out servicing work on water-cooled chillers that involve refrigerant handling.

In June 2022, Singapore ratified the Kigali Amendment to the Montreal Protocol. To meet these obligations, Singapore will phase down the consumption of HFCs by 80 per cent over the next two decades. NEA will continue to work closely with industry stakeholders to support them in reducing HFC consumption, and shift them towards climate-friendly alternatives.

### REDUCING DISPOSABLE CARRIER BAGS

Starting from mid-2023, supermarket operators will be required to charge for disposable carrier bags, to encourage shoppers to adopt more sustainable habits such as bringing their own bags and reducing the use of disposable bags. Supermarket operators with an annual turnover of more than $100 million will need to charge a minimum of five cents for each disposable carrier bag provided at their physical stores. To uphold transparency and accountability, the supermarket operators will also be required to publish information on the number of bags issued, amount of proceeds collected, and how the proceeds are used. The operators are strongly encouraged to use the proceeds to support charitable or environmental causes.

## PROVIDING TIMELY AND RELIABLE WEATHER AND CLIMATE SERVICES

**Situated one degree north of the equator, Singapore's tropical climate presents its own unique set of environmental challenges. To prepare for the effects of climate change, NEA's Meteorological Service Singapore (MSS) provides reliable and responsive weather and climate services, towards greater public safety and a climate resilient economy.**

### IMPROVING WEATHER FORECASTING CAPABILITIES

Climate change and extreme weather changes have resulted in a growing demand for weather and climate prediction information services. To provide improved weather forecasts and alerts for Singapore, MSS will undertake several new initiatives with the latest weather forecasting capabilities and services, underscored by advances in infrastructure, R&D and operations.

These include the development of a MET-Air Traffic Management Integration system to provide actionable high-resolution information for air traffic management. This will improve the efficiency and safety of civil air navigation, and reinforce Singapore's position as a global air hub. In addition, MSS plans to enhance emergency-preparedness at the WOG level, by improving risk assessments for extreme weather events and geohazards such as volcanic eruptions and tsunamis. Beyond this, our strategic initiatives aim to enhance climate predictions to support efforts in areas such as water management, food security and disaster management. These initiatives will enable us to meet increasingly sophisticated requirements for weather-related operations and long-term climate impact assessments.

## GOING FORWARD

NEA's short-term and medium-term focus is to strengthen our climate science capabilities, enhance weather forecasting and hazard prediction capabilities, and provide fit-for-purpose weather services. Key initiatives include conducting the Third National Climate Change Study, which will provide the next generation of localised and high-resolution climate change projections until 2100 for Singapore, driving research under the Climate Science Research Programme Office, and enhancing our Meteorological Observation Systems. Our goal is to observe and understand the weather and climate affecting Singapore, and to provide services to support national needs and international cooperation.

### USING HIGH-PERFORMANCE COMPUTING IN WEATHER AND CLIMATE SCIENCE

To improve the accuracy of our weather and climate prediction capabilities for Singapore and the wider Southeast Asia region, the Centre of Climate Research Singapore is undertaking R&D in numerical climate and weather modelling, as well as machine-learning-based post processing techniques. Priority areas include the development of a 'cSINGV' coupled climate model configuration that can simulate the atmosphere, ocean and land components concurrently, and a higher spatial resolution 'uSINGV' urban climate model configuration representative of Singapore's built-up environment, to provide better realism in capturing small-scale weather and climate processes beyond the current km-scale resolution 'SINGV' MSS climate model.

These scientific developments require significant technological enhancements in our in-house High-performance Computing (HPC) resources, which will be upgraded in 2022 with a new 'HPE Cray EX' supercomputer – with double the computational power of our current system. In addition, we will continue to require significant computational resources from the National Supercomputing Centre's HPC facilities for underpinning weather and climate research, and novel cloud computing infrastructures for climate data analytics.

### STRENGTHENING PARTNERSHIPS IN WEATHER AND CLIMATE SERVICES

As international cooperation is essential to address weather and climate challenges, MSS works to strengthen partnerships with regional and international counterparts, to enhance our capabilities in the provision of meteorological and climate services.

On the regional front, MSS currently serves as the Chair of the ASEAN Sub-Committee on Meteorology and Geophysics (SCMG). This is a three-year term from 2021 to 2023, where we successfully organised the SCMG's 42nd Meeting in November 2021.

As SCMG Chair, we continue to steer the implementation of the 2016 to 2025 Work Plan, to fulfil its objective of enhancing the capabilities of ASEAN Member States in the provision of meteorology and geophysics services.

This includes joint key initiatives spearheaded by Singapore and the ASEAN Specialised Meteorological Centre (ASMC), such as capability-building activities on numerical weather prediction, subseasonal to seasonal predictions, and regional climate change projections. The ASMC, hosted by MSS, has been instrumental in regional capability-building, weather and climate assessments in the region, regional monitoring of land and forest fires, and providing early warnings of transboundary haze.

# VIBRANT HAWKER CULTURE

**Hawker centre dining is truly an integral part of Singapore's culture. As a natural extension of this culture, hawker centres have become time-tested institutions that serve as important places for community bonding.**

## SAFEGUARDING OUR HAWKERS AND THE PUBLIC

NEA played a key role in the nation's fight against COVID-19 by implementing various safe management measures (SMMs) to safeguard hawker centres and markets from emerging COVID-19 cases, protect the hawker workforce by nudging them to be vaccinated and tested regularly, and ensure public safety with safe distancing compliance.

### Keeping Diners Safe

Continuing the efforts from 2020, we stepped up and strengthened safe distancing measures at hawker centres and markets. These measures included the marking of seats and tables to enable safe distancing among patrons and compliance with the group size limits, deploying Safe Distancing Ambassadors to engage and educate the public on safe distancing measures, and enforcement of SMMs by Safe Distancing Enforcement Officers (SDEOs) at hawker centres and markets.

To facilitate contact tracing of COVID-19 cases in the community, temporary fencing was put up at all hawker centres and markets in July 2021, and SafeEntry check-ins were required for access to these premises.

Vaccination-Differentiated Safe Management Measures (VDS) were introduced at hawker centres in October 2021 to allow dining-in by vaccinated persons, while non-vaccinated persons were restricted to take-aways. VDS was further tightened with entry checks at all hawker centres, where visitors had to show their vaccination status before entering the premises. SDEOs were also deployed to conduct checks and take enforcement action against non-compliant patrons.

NEA formed a command-and-control unit to strategise deployments, as well as manage and provide back-office support to ground personnel and outsourced partners. This ensured the smooth operation and execution of the various COVID-19 SMMs and access checks at all hawker centres and markets.

### Protecting our Hawkers

To protect our stakeholders in hawker centres and markets, all stallholders, stall assistants and workers at the centres were required to undergo regular testing as part of the Fast and Easy Testing-Rostered Routine Testing. Stepping up measures in October 2021, unvaccinated individuals were required to be tested twice a week under the Vaccinate or Regular Testing regime. The testing regimes, together with our guidelines on cleaning and disinfection operations for centres exposed to COVID-19 cases, helped to ringfence emerging cases and minimise the risk of transmission. In addition, centres with an upward trend of cases were placed on short-term voluntary closure for deep cleaning and disinfection.

In January 2022, Singapore introduced the Workforce Vaccination Measures (WVM), where only vaccinated individuals were allowed to return to operate the stalls. NEA monitored the compliance of WVM and conducted regular engagements with stakeholders, with reminders to go for vaccination and testing.

## INCULCATING A TRAY AND CROCKERY RETURN CULTURE

In 2021, NEA launched the Clean Tables campaign to galvanise individuals and the community to do their part in keeping public dining places clean and hygienic. In May 2021, we announced that diners will be required to return their used trays, crockery, and clear the litter after their meals from 1 June 2021 at hawker centres. The national Tray and Crockery Return Rate has since seen an increase from an average of 33 per cent prior to the launch of the campaign, to the current average of 88 per cent across hawker centres in Singapore.

Phase 1 of the campaign saw three months of intensive educational efforts to encourage users to be gracious to the next diner by keeping their tables clean. The campaign covered a variety of touchpoints including mass media, outdoor advertising and social media channels. It was supported with on-ground engagement by various outreach partners, government agencies, and relevant NGOs. SG Clean Ambassadors were also deployed at hawker centres to educate and nudge patrons to return their used trays and crockery, and to bin their litter after dining.

Phase 2 involved NEA taking a stronger stance against errant diners. Enforcement officers would first advise diners to clear their used trays, crockery and litter, before taking enforcement action against those who refused to comply.

NEA also introduced several initiatives to make it easier for diners to return their used trays and crockery:

- Installation of additional racks to support the tray and crockery return efforts
- Provision of mobile trolleys to create additional tray and crockery return points when required
- Provision of additional new trays to facilitate the return of used crockery
- Installation of additional netting at racks where applicable, to mitigate bird nuisance

To increase cleaning efficiency, we developed a streamlined table-cleaning workflow for hawker centres in consultation with the Environmental Management Association of Singapore, cleaning companies and cleaners. This resulted in a more sustainable workflow for cleaners who are generally older, by re-organising the deployment of cleaners into two teams – one team would focus on cleaning and sanitising tables, while the other team stationed at the return points would clear the used trays and crockery returned by patrons.

NEA implemented the Clean Tables Support Scheme on 1 May 2021 to inculcate a tray return culture beyond our hawker centres. Under this voluntary co-funding scheme, coffeeshop and food court operators and owners can apply for 50 per cent co-funding, capped at $2,500 per premises, to defray the costs of purchasing trays, as well as putting in place tray and crockery return facilities.

### SUPPORTING HAWKERS THROUGH THE PANDEMIC

To help hawkers tide through the COVID-19 pandemic, the Government provided various COVID-19 relief measures to stallholders in 2020 and 2021. These included 10 months of rental waivers and six months of subsidies for table-cleaning and centralised dishwashing services, and a one-off cash payout of $500 through the Market and Hawker Centre Relief Fund.

In early 2022, the Government announced the Small Business Recovery Grant to support small and medium enterprises that have been most affected by the COVID-19 restrictions, including hawker centre and market stallholders licensed by the Singapore Food Agency (SFA). Eligible stallholders have received the payouts in batches from June 2022.

To help hawkers pivot their businesses and embrace digital sales channels, various stakeholders, such as hawkers, community groups and food delivery platforms, came together to form the 'SG Together Alliance for Action – Online Ordering For Hawkers' initiative. As an added incentive, hawkers who onboarded online food delivery platforms could claim a one-time assistance of $500 to defray their onboarding costs by December 2021.

## GOING FORWARD

NEA's mission is to develop and manage hawker centres as vibrant communal spaces, offering a variety of affordable cooked food options in a clean and safe environment, and ensure a sustainable hawker trade.

Our short-term focus is to take steps to strengthen the resilience of our hawker centres and markets against future public health threats. We seek to ensure business continuity at our hawker centres by taking the appropriate health and hygiene measures, and creating a clean and hygienic environment for hawkers and patrons.

NEA is also focused on developing new hawker centres and rejuvenating existing centres, enhancing the vibrancy and productivity of hawker centres, and ensuring the sustainability of the hawker trade. With the successful inscription of Hawker Culture on UNESCO's Representative List of Intangible Cultural Heritage of Humanity, these efforts will go towards promoting and sustaining a vibrant Hawker Culture in the long term.

### GROWING THE HAWKER CENTRE PRESENCE

With more than 100 hawker centres in Singapore, patrons have access to a good mix of affordable cooked food options. These include seven new hawker centres at Bukit Panjang, Hougang (Ci Yuan), Woodlands (Kampung Admiralty), Jurong West, Tampines (Our Tampines Hub), Pasir Ris Central and Yishun Park. Other new hawker centres that have been announced are located at Sembawang (Bukit Canberra), Bidadari (Woodleigh Village), Bukit Batok West, Bukit Panjang North (Senja), Sengkang (Fernvale, Buangkok and Anchorvale Village), Punggol (One Punggol and Punggol Digital District) and Choa Chu Kang (Yew Tee and Town Centre).

The status of the upcoming new and replacement hawker centres are as follows:

**New Hawker Centres – Under Construction**

- Bukit Canberra Hawker Centre, Sembawang
  - BCA Green Mark Platinum (target)
  - BCA Universal Design Mark Gold$^{PLUS}$ (Design)
- Buangkok Hawker Centre, Sengkang
  - BCA Green Mark Gold$^{PLUS}$
  - BCA Universal Design Mark Gold$^{PLUS}$ (Design)
- Woodleigh Village Hawker Centre, Bidadari
  - BCA Green Mark Gold$^{PLUS}$
- Anchorvale Village Hawker Centre, Sengkang
  - BCA Green Mark Platinum
- Bukit Batok West Hawker Centre
  - BCA Green Mark Gold (target)
  - BCA Universal Design Mark Gold (target)
- Punggol Digital District Hawker Centre
  - BCA Green Mark Platinum (target)
  - BCA Universal Design Mark Platinum (target)

**New Hawker Centres – Completed Construction**

- Fernvale Hawker Centre & Market
  - BCA Green Mark Platinum
  - BCA Universal Design Mark Gold$^{PLUS}$ (Design)
- One Punggol Hawker Centre
  - BCA Green Mark Platinum
  - BCA Universal Design (Platinum)
- Senja Hawker Centre, Bukit Panjang North
  - BCA Green Mark Platinum
  - BCA Universal Design Mark Gold (Design)

**New Replacement Centres – Under Construction**

- Telok Blangah Hawker Centre and Market
  - BCA Green Mark Platinum
  - Replaces Telok Blangah Drive Market & Food Centre

**New Replacement Centres – Completed Construction**

- Market Street Hawker Centre
  - BCA Green Mark Platinum
  - BCA Universal Design Mark Gold$^{PLUS}$ (Design)
- Margaret Drive Hawker Centre, Dawson C3
  - BCA Green Mark certified
  - Replaces Commonwealth Drive Food Centre

# Together with Our Partners and the Community

## ENVIRONMENTAL SERVICES INDUSTRY AS A FORCE MULTIPLIER FOR MISSION SUCCESS

**Singapore's commitment to environmental sustainability and economic growth is reflected in the Environmental Services Industry Transformation Map developed by NEA to drive change in the industry. We work closely with our stakeholders to form strategic partnerships, and recognise their sustainability efforts through awards and accolades.**

### TRANSFORMING THE INDUSTRY WITH ROBOTICS

NEA encourages the greater use of technology and automation to transform the Environmental Services industry and create higher-value jobs. Together with our partners, we are developing robotic solutions for the cleaning of public spaces, waste management, as well as inspection, monitoring and sampling functions.

Following the successful completion of public road trials in 2021, the Autonomous Environmental Service Vehicle (AESV) for road cleaning is into its second phase, where fleet management and cleanliness monitoring functions are being developed. In particular, the tele-operation of the AESV – which empowers users to control the vehicle remotely – has captured significant attention, especially in other industries where the nature of the work is dangerous and repetitive, such as heavy lifting activities or crane operations.

NEA has since expanded our Environmental Robotics R&D Programme beyond road cleaning. These include other applications such as pavement sweeping, closed drain and false ceiling inspection, and anomalies detection in niches.

### ENCOURAGING FOOD RESOURCE VALORISATION

NEA launched the inaugural Food Resource Valorisation Awards (FVA) to recognise the efforts of companies engaging in food resource valorisation, and raise awareness of the food resource valorisation concept. While conventional food waste treatment processes food waste into products such as compost or non-potable water, food resource valorisation converts food waste, such as homogenous by-products, rejects and mixed food waste, into higher value products that contribute to a sustainable economy.

At the FVA 2021 event held on 29 September 2021, 11 winners across two award categories were recognised for their outstanding efforts in driving sustainability and closing the food resource loop through valorisation solutions. The Merit Award category is for projects operating on a commercial scale for a minimum of two years, while the Participation Award category recognises projects in the pilot or R&D phase.

### LEADING GLOBAL SUSTAINABLE DEVELOPMENT DISCUSSIONS

NEA regularly engages leading international experts and various industry stakeholders on climate change resilience, sustainable urban development, and environmental issues.

In partnership with Temasek's Ecosperity Week 2021, we organised the CleanEnviro Summit Singapore (CESG) Catalyst 2021 event on 29 September 2021. Themed 'Enhancing Sustainable Development in a Pandemic World', the hybrid event convened more than 200 policymakers and leaders from the business, government and international institutions to discuss and share insights on a strategic blueprint for climate-resilient cities, and how cities can build resilience against public health threats.

These conversations continued at CESG 2022 in April 2022, to explore sustainable solutions for environmental issues for a better tomorrow.

### DEEPENING SUSTAINABILITY COLLABORATIONS

On 20 October 2021, NEA signed a Memorandum of Understanding with the US-ASEAN Business Council to strengthen cooperation on sustainability. The areas of collaboration include waste resource management and recycling, addressing marine litter through the management of land-based sources of pollution, exchanges on carbon credits, and building a circular economy with sustainable supply chains. As part of the ongoing collaborations, we supported the US-ASEAN Business Council Webinar as one of the panellists on Challenges and Opportunities for the Circular Economy in ASEAN on 16 December 2021. In turn, the US-ASEAN Business Council supported CESG 2022 as a Strategic Partner, from 17 to 21 April 2022.

### STANDARDISING RENEWABLE ENERGY CERTIFICATES

With growing climate awareness, Renewable Energy Certificates (REC) have become a common means for energy users, including businesses, to fulfil their decarbonisation commitments. Through the purchase of RECs, users can indirectly source electricity from renewable sources.

To facilitate consistency for the transaction and management of RECs, the Singapore Standards Council and ESG, together with NEA, the Energy Market Authority, Sustainable Energy Association of Singapore and industry players, developed the Singapore Standard (SS) 673: Code of Practice for Renewable Energy Certificates.

The SS 673 provides a clear framework to improve the integrity of measurement, reporting and verification requirements for the issuance and management of RECs. It covers guidelines across the life cycle of RECs – from production, tracking and management, to the usage of the certificates for renewable energy claims in Singapore. The SS 673 is applicable to renewable energy installation owners, REC registries, REC intermediaries, traders, brokers, third-party verifiers, issuers, and end-users.

## STAKEHOLDER ENGAGEMENT HIGHLIGHTS

### ENGAGING THE INDUSTRY ON NEW RADIOLOGICAL SECURITY GUIDELINES

In 2021, NEA developed new guidelines to strengthen the security of radioactive materials in three areas – the improvement of security measures at radioactive material storage sites, the enhancement of security of mobile devices containing radioactive material, and in inventory control and stock keeping.

To understand industry readiness in implementing these measures, NEA conducted industry consultations in April and May 2022, and will work with the industry to implement the security enhancements based on the new guidelines.

### ENGAGING THE FUNERAL INDUSTRY

To maintain high public health standards when funeral services providers handle the COVID-19 deceased in the community, NEA nudged the funeral industry to be vaccinated, tested regularly, and trained in the Basic Infectious Control Course. Safe management measures were also introduced at government-managed after-death facilities, funeral parlours and funeral wakes.

In 2021, NEA organised a series of briefing and training sessions to prepare funeral services providers for the implementation of the digital death certificate. To increase the funeral industry's operational readiness, accord dignity to the deceased, and ease the transition for bereaved family members, we developed online materials and made more information readily available on the NEA website. NEA will continue to work closely with our stakeholders and engage religious organisations, to improve the quality of after-death facilities and services in Singapore.

## ACTIVE COMMUNITY AS STEWARDS OF THE ENVIRONMENT

**NEA works closely with the People, Public and Private sectors to drive environmental ownership through various campaigns and programmes. These sectors are critical to Singapore's journey towards sustainability, as everyone can contribute to a cleaner and greener way of life.**

### CULTIVATING YOUNG ENVIRONMENTAL LEADERS

In July 2021, NEA launched the Youth for Environmental Sustainability (YES) programme, to provide a robust well-rounded platform for youths to develop environmental leadership capabilities.

To engage youths effectively during the COVID-19 pandemic, the YES programme rolled out online content in collaboration with popular content publishers, and launched a #SGEcoHacks challenge with prizes sponsored by Accor Hotels & Resorts. Beyond the online efforts, we also cemented a new partnership with the Youth Corps Singapore, through NEA's YES edition of the Youth Corps Leaders Programme. Designed as a capability development programme for aspiring youth leaders, the programme comprises a structured learning framework coupled with a service-learning project, to provide them with environmental sustainability knowledge, skillsets and experience. As part of YES, NEA also partnered organisations such as L'Oréal Group Singapore, the National University of Singapore (NUS), SembWaste, and the South West Community Development Council to implement various initiatives to support youth actions.

### KEEPING PRIVATE ESTATE ROADS CLEAN

Vehicles parked along private estate roads hinder the cleaning efficiency of mechanical road sweepers. With a common goal towards cleaner public streets, NEA engaged residents of private estates to implement alternative roadside parking initiatives on planned road sweeping dates. We sought their cooperation to refrain from parking their vehicles on one side of the road on a specified day and time, after which they could resume their normal parking arrangements. To ensure clear communication, we placed signs and distributed pamphlets to inform residents in the estates ahead of the planned road sweeping dates.

Following the successful implementation in six residential estates, the Alternate Parking Project was extended to Jalan Wajek, a residential estate in the Northwest region, where it was warmly received. It will be extended to five more estates by December 2022.

## STAKEHOLDER ENGAGEMENT HIGHLIGHTS

### ENCOURAGING RECYCLING WITH BEHAVIOURAL INSIGHTS

To broaden recycling efforts, NEA collaborated with NUS to trial specially designed 'Recycle Right' recycling bins in public spaces. These bins were developed by NUS and piloted on the university campus, nudging users to recycle correctly and thereby reduce contamination. Building on the encouraging results from the campus pilot, NEA collaborated with NUS to conduct a trial at three shopping malls from November 2021 to January 2022. Data collected showed improvements in user recycling behaviour and a reduction in contamination of the recycling bins. In addition, some premises owners have expressed interest in using these recycling bins on their premises to aid their recycling activities.

## PUBLIC SECTOR TAKING THE LEAD IN ENVIRONMENTAL SUSTAINABILITY

**Public agencies in Singapore often come together to coordinate operations, share expertise and resources, and collaborate in inter-agency committees and task forces. To harness this synergy, NEA partners fellow public agencies in programmes to drive sustainability across the public sector.**

### BUILDING A COMMUNITY OF PRACTICE IN RADIOCHEMISTRY

NEA monitors ambient radiation levels for any increases in radioactive substances in the environment, food and water. We also work with regional regulators and ensure Singapore's compliance with international obligations for nuclear safety.

We have progressively ramped up our capabilities in ambient radioactivity monitoring and testing. In September 2021, NEA's National Radiochemistry Laboratory championed a Community of Practice (COP) for Radiochemistry Laboratories, with members from

the Home Team Science & Technology Agency, PUB and SFA. The COP aims to advance technical knowledge and competency in radiochemistry through innovation, promote an open and collaborative community, sustain capability during peacetime, and be able to ramp up capacities to meet a surge in demand during emergencies.

Since its formation, the COP has conducted several activities of common interests, including sharing sessions on Proficiency Test Performance, and Determination of Gross Alpha and Gross Beta in water samples. Moving forward, we will continue to work with COP members to build capabilities and deepen collaboration.

### INTEGRATING SENSORS FOR BETTER AIR QUALITY

NEA collaborated with the Government Technology Agency of Singapore to leverage the Singapore Government Tech Stack for Device Management Control and Data Acquisition to implement new sensor system projects. The collaboration successfully shortened the onboarding process of new sensors to strengthen capabilities in chemical pollution monitoring and control. This allows the operations team to easily access near real-time sensor readings and alerts through an existing secured platform that is compliant with stringent security policies and standards. We will progressively onboard other sensor systems that are due for a technology refresh, while keeping abreast of the latest technological innovations in air quality monitoring.

## STAKEHOLDER ENGAGEMENT HIGHLIGHTS

### ACCELERATING OPERATIONAL READINESS

The COVID-19 pandemic highlighted the need for the public sector to be ready and prepared at all times, to respond swiftly and effectively to any dynamic ground situations, such as environmental and public health incidents.

In collaboration with WOG agencies, a series of proofs-of-concept have been earmarked to drive data-centric operations through the proliferation of data. This includes exploring industrial Internet of Things convergence sensor devices with IT applications for more seamless sensor operations, to training video analytics models on a one-stop platform as part of a bigger system.

## INTERNATIONAL PARTNERSHIPS AS STRATEGIC LEVERS

**The effects of climate change further intensify Singapore's vulnerability to transboundary environmental threats. To foster collaboration and knowledge exchange on the global stage, NEA engages our ASEAN and international counterparts, and leverages bilateral and multilateral platforms to promote Singapore's expertise in environmental management beyond our shores.**

### DEVELOPING EMERGENCY RESPONSE PREPAREDNESS

A nuclear or radiological emergency in Southeast Asia or outside the ASEAN region could have far-reaching effects on ASEAN countries, due to trans-boundary effects of radioactive plume and the movement of people, goods and services.

As a member of the ASEAN Network of Regulatory Bodies on Atomic Energy (ASEANTOM), we spearheaded the development of the regional technical cooperation project concept with the International Atomic Energy Agency to support capability building in nuclear emergency preparedness and response (EPR) for ASEAN Member States. The key deliverable of this regional project was the development of the ASEAN Protocol for Preparedness and Response to a Nuclear or Radiological Emergency.

From 2018 to 2021, we worked on drafting the ASEAN Protocol for submission to ASEANTOM for review, as part of the working group with Thailand, Malaysia and Cambodia. The ASEAN Protocol was officially endorsed at the ASEAN Senior Officials' Meeting in October 2021.

The ASEAN Protocol is a high-level document to frame future EPR arrangements and cooperation for radiological or nuclear emergencies. It serves as the overarching framework to harmonise decision-making and public communications in the ASEAN region during nuclear or radiological emergencies.

### COMBATING MARINE DEBRIS

NEA worked closely with fellow ASEAN Member States under the ASEAN Working Group on Coastal and Marine Environment to develop the ASEAN Regional Action Plan for Combating Marine Debris. The Plan, which followed up on ASEAN's commitments in the Bangkok Declaration on Combating Marine Debris in the ASEAN Region, outlined an integrated approach to addressing marine plastic pollution through 14 regional actions

targeted at three key stages of the plastics value chain. The Plan, which will be implemented between 2021 and 2025, was launched at a virtual ceremony on 28 May 2021.

### ADDRESSING REGIONAL MARINE PLASTIC DEBRIS

NEA co-organised the East Asia Summit Workshop on Marine Plastic on 14 and 15 February 2022, in partnership with Australia's Commonwealth Scientific and Industrial Research Organisation and India's National Centre for Coastal Research. The virtual workshop featured 15 speakers from 11 countries, who shared the latest research, best practices, and technology in addressing marine plastic debris. We presented how Singapore's Zero Waste initiatives helped reduce and eliminate plastic pollution in Singapore. The event received the support and participation of more than 100 participants from the 18 Participating Countries.

### FORTIFYING REGIONAL CAPABILITIES

To facilitate regional knowledge exchange and capability building, we organised two training programmes on Urban Solid Waste Management and Air Quality Management for the ASEAN Member States (AMS) in November 2020 and January 2022 respectively.

Organised under the auspices of the ASEAN Working Group on Chemicals and Waste chaired by Singapore, the training programmes equipped senior government officials with a holistic understanding and overview of the conceptual frameworks.

The two workshops were well-attended with 53 participants from the AMS, who gained insights on the environmental challenges arising from climate change and Singapore's strategies in urban environment development, as well as each country's unique challenges and experiences.

## STAKEHOLDER ENGAGEMENT HIGHLIGHTS

### FOSTERING BILATERAL TIES VIRTUALLY

Although the COVID-19 travel restrictions prevented the exchange of physical visits in 2021, we continued to engage our foreign counterparts through virtual platforms and strengthened partnerships towards shared goals. These initiatives included institutionalised meetings, workshops and training programmes, as well as ad-hoc professional and technical exchanges of information. Through these efforts, we expanded bilateral cooperation on environmental matters with strategic partners, including Australia, Brunei Darussalam, China, Republic of Korea, the Netherlands, and the United States of America.

# Enabled by a High Performance and Future-ready NEA

## CREATION OF CAPACITY

**NEA is constantly innovating to create capacity and achieve process excellence. As the demand for environmental services rises, it is essential to shift towards a pro-enterprise mindset through innovative solutions across regulations and practices, to ensure a safe and healthy environment for all employees.**

### DRIVING INNOVATION EXCELLENCE IN NEA

To cultivate and sustain an innovation-driven culture, NEA has an innovation framework based on the tenets of organisation development and behavioural sciences. The framework pulls together six levers to drive innovation excellence for value creation.

NEA's Innovation Framework




### RECOGNISING INNOVATIVE EFFORTS

Encouraging innovation is our way of inspiring staff to keep thinking out of the box. In 2021, 32 NEA innovation awards were presented to recognise individuals and project teams for their innovative spirit and solutions.

NEA was also awarded three certificates – two Par Excellence and one Excellence – at the International Convention on Quality Control Circles, for optimising the deployment of safe distancing officers, innovating a microscope for officers to identify mosquito species on the field, and reducing the time required to track inspections at construction sites. These awards affirm our commitment to innovation, and reflect our innovation-driven culture.

## STRONG CAPABILITIES FOR MISSION SUCCESS

**To harness data analytics effectively, NEA works in tandem with a network of relevant partners to advance our capabilities through R&D. With the aim of optimising regulations and strengthening our collective science, technology and engineering resources, we strive to ensure win-win solutions for all stakeholders.**

### DEVELOPING AUTOMATED TRACKING FOR COVID-19 TESTING

The pandemic saw the implementation of various safe management measures nationwide, such as Fast and Easy Testing-Rostered Routine Testing, Vaccinate or Regular Testing, and Workforce Vaccination Measures. To cope with the mandatory requirements, we needed an efficient way to schedule and track the appointments of over 20,000 individuals, and monitor their compliance to the measures. Through the use of data analytics and robotic process automation, NEA developed effective and resource efficient solutions that allowed us to gain a better overview of the situation, ensure compliance, and augment staff capacity to handle a surge in workload. This resulted in 42,977 man-hour savings from September 2021 to February 2022.

### PROVIDING REAL-TIME ASSISTANCE FOR DENGUE

'Ask Jamie' has been an invaluable Virtual Assistant on our website, providing users with responses to frequently asked questions. In July 2021, we integrated the Dengue Application Programming Interface to give 'Jamie' immediate access to the dengue situation, enabling her to answer questions users have on specific locations. This helped to improve online user experience, and enabled users to take the necessary precautions to avoid dengue clusters and keep dengue at bay.

### DIGITALISING OUR OPERATIONS

Aligned with the targets in the Digital Government Blueprint, we are digitalising our services and operations for greater efficiency and public responsiveness. NEA has made improvements in four key areas to improve user experience, thereby achieving 100 per cent e-payment for outbound services. We are on target to achieve 100 per cent e-payment for inbound services by 2023. Additionally, 78 per cent of the transactions can be completed end-to-end digitally, marking an increase of 30 per cent from 2020. To improve user experience, 66 per cent of our services now enable the pre-filled government data function for greater convenience. We are also able to offer a digital signature option in 29 per cent of our services, and will continue to introduce this in other services.

### ONBOARDING THE COMMERCIAL CLOUD

In 2021, we made significant progress with onboarding 65 per cent of our eligible systems onto the Commercial Cloud, marking an increase of 18 per cent from 2020. This was achieved by adopting modern cloud application design practices, and tapping on cloud native services to maximise the benefits.

### HARNESSING DATA AND ANALYTICS

NEA continually explores ways to improve service delivery and work processes with data and analytics. In 2021, the Data Analytics Community of Practice was formed to grow and develop a competent community of analytics enthusiasts and practitioners within NEA. 'Brown Bag Sessions' were organised to impart quick tips, tools and concepts related to analytics for application in daily work. A data workshop was also organised to share analytics use cases, and provide a platform for staff to brainstorm potential analytics projects.

These efforts brought about a higher participation rate in the annual Analytics Drive competition, which aimed to encourage the use of analytics to provide better situational awareness, gain useful insights for better decision making, and enhance operational effectiveness. In 2021, the competition saw 28 project submissions – the highest since its inception, signifying an increase in analytics interest across the organisation.

## STRONG CORPORATE PRACTICES FOR A SUSTAINABLE ORGANISATION

**It is crucial for NEA to build robust capabilities, given our diverse scope of work and wide-ranging stakeholders who depend on our policies and services. Continuously improving our internal capabilities allows us to respond swiftly and effectively in today's complex environment.**

### FOSTERING A SENSE OF BELONGING

People are central to NEA's organisational performance and excellence. NEA continues to foster a sense of belonging and promote the development and professionalism of our people.

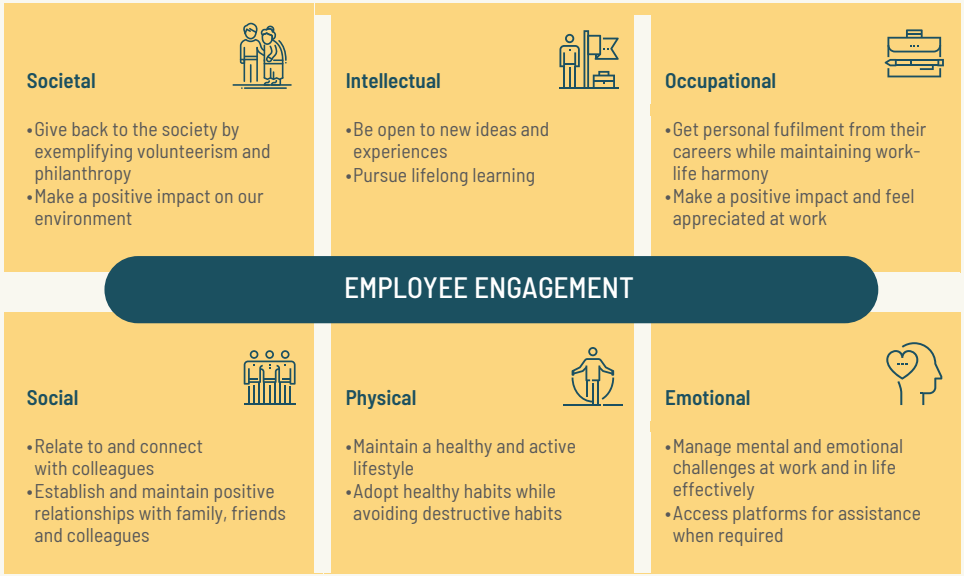
To achieve this, NEA has five long-term human resource strategies:

- Brand NEA to attract the right talent
- Develop workforce capabilities
- Build an engaged and resilient workforce
- Grow leadership pipeline
- Improve human resource processes and systems

### DRIVING EMPLOYEE ENGAGEMENT

NEA's Employee Engagement (EE) Framework aims to engage staff beyond their core work by tailoring to their different needs through the EE Framework. The framework is also aligned with the Public Service Cares initiative that supports the Singapore Cares (SG Cares) movement to build a caring and inclusive society.

Employee Engagement Framework




The dimensions of the EE framework are led by respective divisions or departments, and entities for greater focus and synergy.

| | |
|---|---|
| Human Resource and Organisation Development Division<br>Captain Green Cares Community (Care for Staff and Mental Wellness) | Occupational, Emotional |
| NEA Sports and Social Committee | Social, Physical |
| Captain Green Cares Community (Staff Volunteerism and Philanthropy)<br>Sustainability Division (Environmental Sustainability) | Societal |
| Corporate Excellence Department (Innovation)<br>Singapore Environment Institute (Lifelong Learning) | Intellectual |

## CREATING A POSITIVE WORK ENVIRONMENT

NEA creates a positive work environment for our employees through fair employment and work-life effectiveness. Our management is actively involved in setting the vision and direction, policy formulation, implementation and feedback loop for fair employment practices. We signed the Employers' Pledge of Fair Employment Practices in April 2007, to review employment practices regularly in order to align with the fair employment principles set out by the Tripartite Alliance for Fair Employment Practices. NEA's terms of employment are also guided by Singapore's legislations, including the Employment Act 1968, Workplace Safety and Health Act 2006, and the Retirement and Re-employment Act 1993.

NEA's journey towards a future-ready workforce led to the development of a set of Cultural Capabilities, which guide our staff in acquiring desirable traits to cope with the transformation.

### Inclusive Hiring

NEA is proud to support the hiring of persons with disabilities as part of our inclusive workplace strategy. We have put in place a holistic framework covering strategies, initiatives and action plans to champion inclusive hiring. We also support SG Enable in its career fairs and internship programmes. As of 31 March 2022, we have helped create a conducive workplace for 21 persons with disabilities.

### Unions

Our people are our greatest assets, and we work closely with the Amalgamated Union of Public Employees (AUPE) and the Amalgamated Union of Public Employees - Daily-Rated Workers Branch (AUPE-DRW Branch) to ensure we take care of them.

### Employee Grievance

NEA has put in place a grievance handling framework, and is committed to working closely with the union to address and resolve staff feedback and concerns expeditiously.

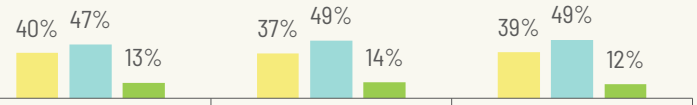
## Staff Demographics

As of 31 March 2022, NEA had a diverse workforce of 3,799 working in various locations islandwide.

Breakdown by Age

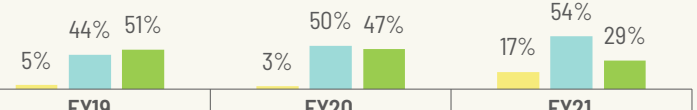
NEA continued to recruit employees of all ages, with those between the ages of 30 to 50 years old leading the group.

**No. of Employees in NEA**



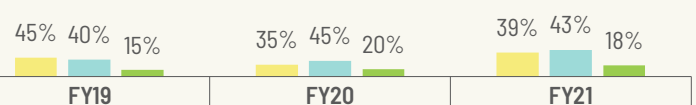

| | FY19 | FY20 | FY21 |
|---|---|---|---|
| > 50 years old | 1,444 | 1,426 | 1,470 |
| 30-50 years old | 1,719 | 1,845 | 1,870 |
| < 30 years old | 491 | 520 | 459 |

**No. of Employees Recruited**




| | FY19 | FY20 | FY21 |
|---|---|---|---|
| > 50 years old | 19 | 13 | 82 |
| 30-50 years old | 164 | 207 | 255 |
| < 30 years old | 189 | 191 | 138 |

**No. of Employees Who Left NEA**




| | FY19 | FY20 | FY21 |
|---|---|---|---|
| > 50 years old | 150 | 95 | 188 |
| 30-50 years old | 132 | 123 | 209 |
| < 30 years old | 51 | 56 | 87 |

Breakdown by Gender

Due to the nature of NEA's operational work which is skewed towards enforcement and field operations, there are more male applicants and hence higher chances of a male assuming the post. Nevertheless, both genders are given equal consideration in the recruitment process.

**No. of Employees in NEA**

| | FY19 | FY20 | FY21 |
|---|---|---|---|
| | 34% | 35% | 35% |
| | 66% | 65% | 65% |
| Female | 1,241 | 1,309 | 1,314 |
| Male | 2,413 | 2,482 | 2,485 |

**No. of Employees Recruited**

| | FY19 | FY20 | FY21 |
|---|---|---|---|
| | 40% | 40% | 33% |
| | 60% | 60% | 67% |
| Female | 147 | 165 | 159 |
| Male | 225 | 246 | 316 |

**No. of Employees Who Left NEA**

| | FY19 | FY20 | FY21 |
|---|---|---|---|
| | 29% | 35% | 33% |
| | 71% | 65% | 67% |
| Female | 98 | 97 | 160 |
| Male | 235 | 177 | 324 |

Breakdown by Employment Type

The majority of our employees are employed on a full-time basis. During COVID-19, Safe Distancing Ambassadors (SDA) were the most common type of workers who were not NEA's employees. These SDAs were engaged to conduct safe management and engagement operations at NEA-managed hawker centres and markets. They were engaged through recruitment agencies appointed under VITAL, or via business-to-business contracts with the private sector for the redeployment of their staff to NEA.

**No. of Non-employees**

| | FY20 | FY21 |
|---|---|---|
| SDAs | 399 (77%) | 273 (82%) |
| Others | 120 (23%) | 60 (18%) |

Compared to FY2020, the total number of SDAs in the current reporting period had declined as Singapore progressed towards endemicity. Apart from the SDAs, the number of workers engaged for other reasons had also declined by more than 40 per cent, primarily due to the implementation of a tightened criteria in May 2021 for the engagement of such workers.

### Support for Senior Employees

We value our workers regardless of age and have actively re-employed our eligible retirees. A large proportion of our employees are older workers, of which 39 per cent are aged 50 and above. Employees who were ineligible to be offered re-employment were given an Employee Assistance Payment. We also connect them with organisations such as the Employment and Employability Institute, which offers employment and training assistance.

### Employee Benefits

We value our employees and strive to provide full-time or part-time employees with the same set of benefits. Temporary staff hired via outsourced agencies are covered by their respective agency's benefits policy.

All NEA staff are covered under the Group Personal Accident (GPA) Insurance and Work Injury Compensation (WIC) Insurance. The GPA insurance provides employees with a $100,000 24-hour worldwide coverage for death and permanent disablement due to accidental means beyond work, while the WIC insurance provides coverage for work-related injuries or death as per the Work Injury Compensation Act 2019.

Staff also have the option to take up additional coverage for themselves, their spouses and eligible children under the Public Officers Group Insurance Scheme. They can decide on the insurance coverage and amount, and purchase this directly from Singlife at special rates for Public Service officers.

### Family-friendly Human Resource Policies

We support our staff in their parenthood journey by providing paid parental leave to both male and female employees. Doing so aims to encourage shared parental responsibilities and strengthen family bonds.

Our human resource policies support the career development of employees with families, and affirm our continued commitment to equality and diversity in the workplace. This has paid off with a high retention rate among male and female employees who took parental leave in FY2021. Among the employees who returned to work after their parental leave ended, 94 per cent were still employed 12 months later.

**Return-to-work and Retention Rates of Employees Who Took Parental Leave**

| | Male | | | Female | | |
|---|---|---|---|---|---|---|
| | FY19 | FY20 | FY21 | FY19 | FY20 | FY21 |
| Return-to-work Rate[12] | 100% | 99% | 100% | 96% | 100% | 100% |
| Retention Rate[13] | 99% | 95% | 83% | 90% | 95% | 84% |

**No. of Employees Who Took Parental Leave**

| | FY19 | FY20 | FY21 |
|---|---|---|---|
| Female | 52 | 56 | 51 |
| Male | 85 | 111 | 98 |

**No. of Employees Who Returned to Work After Taking Parental Leave**

| | FY19 | FY20 | FY21 |
|---|---|---|---|
| Female | 50 | 56 | 51 |
| Male | 85 | 110 | 98 |

[12] *The 'return-to-work-rate' is defined as the percentage of employees who returned to work after parental leave in comparison against the total number of employees due to return to work after taking parental leave*

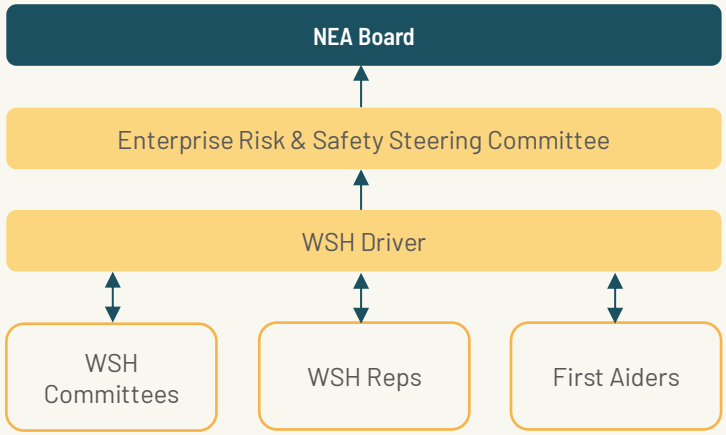
[13] *The 'retention rate' is defined as the percentage of employees retained 12 months after returning from parental leave in the previous FY*

## PROMOTING WORKPLACE SAFETY AND HEALTH

Workplace safety and health (WSH) is a top priority for NEA. WSH matters are reviewed and managed by a network of stakeholders across the organisation, who report to the Enterprise Risk and Safety Steering Committee for strategic guidance.

To promote a strong safety culture under the WSH Governance Structure, we actively monitor our WSH performance, systematically review and address safety risks, and raise awareness of good practices.

### WSH Governance Structure




In 2021, 40 workplace injuries were reported, four of which were sustained by workers who were not NEA employees. The majority of cases were minor injuries, with the most common incident type being 'Slips, Trips and Falls'. All incidents were investigated, with learning lessons shared across the organisation in a timely manner.

In September 2021, a localised explosion in the electrical switch room of the Tuas Incineration Plant resulted in two workplace fatalities and one major injury. We provided support to the affected families, and worked closely with MOM, the Singapore Civil Defence Force, and other technical agencies to investigate the incident. The team also conducted extensive checks on the machinery and reviewed safety procedures before resuming partial operations in October 2021. NEA also stepped up emphasis on workplace safety across all our sites.

### Employee Workplace injuries in 2021

| | No. of Injuries | Injury Rate per 100,000 workers |
|---|---|---|
| No. of Fatal Injuries | 2 | 51.4 |
| No. of Major Injuries | 1 | 25.7 |
| No. of Minor Injuries | 33 | 847.9 |

Apart from monitoring workplace injuries, NEA incentivises holistic WSH performance through internal WSH Scoreboard Awards for low incident rates, strong WSH programmes and innovation, and safe work practices. Safety briefings and trainings are also regularly conducted to raise awareness of work-related hazards and safety incidents.

## STRENGTHENING CORPORATE PRACTICES AND GOVERNANCE

NEA has put in place best practices to ensure continued transparency and accountability to stakeholders. Other government agencies, such as the Ministry of Finance and the Auditor-General's Office, serve as external controls to reinforce ethical and responsible conduct within NEA.

### Pursuing Sustainability in Procurement Practices

As part of our sustainable procurement practices, we encourage suppliers to follow our criteria in integrating sustainability practices in their work processes.

In FY2021, approximately 215 suppliers were awarded contracts to support NEA's operations. In compliance with the Instruction Manual on Procurement, NEA ensures three principles – transparency, open and fair competition, and value for money – are met during the procurement process.

NEA established the Price Quality Scoring Framework for consistent and objective evaluation of bids submitted by suppliers. All open quotations and tenders are posted on a publicly available system – GeBIZ – for suppliers to submit their bids. To avoid conflict of interest, the approving authority for awarding the quotations or tenders cannot be from the procuring division. In FY2021, NEA also improved our procurement processes through digitisation and robotic process automation, for greater resource optimisation and sustainability.

### Ensuring Accountability and Resource Optimisation

NEA was allocated an operating grant of $1,086 million for FY2021, mainly used for the implementation of key programmes to align with the environmental priorities and desired outcomes outlined in NEA's strategy map.

Major NEA projects funded include:

- Development of crematorium, cemetery and related works ($35 million)
- Development of hawker centres ($19 million)
- Development of waste management facilities ($5 million)

**FY2021 Operating Expenditure**

| Domain Area | Amount ($m) |
|---|---|
| High Public Health Standards | 347 |
| Safe, Healthy and Conducive Living Environment | 78 |
| Sustainable and Resource Efficient Singapore | 295 |
| Timely and Reliable Weather and Climate Services | 46 |
| Vibrant Hawker Culture | 106 |
| High Performance and Future-ready NEA | 256 |
| **Total Expenditure (based on audited financial statements)** | 1,128 |

### Ensuring Financial Sustainability

NEA is funded mainly by government grants. It is vital that NEA makes financially responsible and sustainable decisions for daily operations. NEA puts in place the following practices to ensure financial sustainability:

Regular Review of Fees and Charges

Fees and charges are reviewed periodically and are set based on policy intent. The principles and policy considerations when setting fees and charges are:

- Recovering costs directly from the users of that service
- Accounting for the full cost incurred in providing a good or service
- Ensuring that NEA does not compete with the private sector in the provision of goods and services
- Keeping costs affordable and fees low, by striving to improve business processes

Continual Review of Financial Policies

To ensure resource optimisation and to uphold good corporate governance, NEA sets prudent financial policies and procedures, and reviews them regularly. These policies and procedures are published on the intranet to ensure compliance as well as to standardise departments' practices.

Optimal Budget and Resource Allocation

The budget for the new FY is approved by the NEA Board before the start of each FY. As fiscal resources are limited, the budget must be allocated optimally among departments. To ensure accountability and responsible budgeting, a budget utilisation rate of 95 per cent is set as a target for departments to achieve. Periodic performance reviews against budget allocation are conducted with departments, and reported at management meetings regularly.

## CONDUCIVE CULTURE FOR COLLABORATION, LEARNING AND INNOVATION

**NEA advocates fair employment, career opportunities, and support for work-life effectiveness. We are also committed to equip our employees with skills, knowledge and competencies to perform current and future roles.**

### SUPPORTING EMPLOYEE DEVELOPMENT

Employee development at NEA is supported by a robust learning infrastructure and senior management. This enables the smooth and seamless execution of learning strategies. Before the start of each year, all employees are encouraged to discuss their career aspirations with their supervisors during the performance management review. Employees then set career goals and development plans that are aligned with NEA's objectives.

In FY2021, almost all employees completed their annual performance management review. NEA supports the career development plans set during the performance management review through available programmes under the learning infrastructure for employee development.

**Completion Rate of Performance Management Review**

| | FY19 | FY20 | FY21 |
|---|---|---|---|
| Male | 98.2% | 98.6% | 92.2% |
| Female | 97.3% | 98.8% | 90.1% |

A comprehensive range of initiatives have been put in place to develop NEA employees.

**Developmental Initiatives for NEA Employees in FY2021**

| Initiatives | Objectives | Outcomes |
|---|---|---|
| Enhancement of Posting Framework | Provides more structured posting plans for staff with three postings within the first 10 years, and two postings in the next 10 years | About 12.6% of NEA's promising staff, who had spent three or more years on the same job, were posted to different portfolios |
| Internal Market Place (IMP) | Provides staff the opportunity to fill internal positions and develop their career within NEA – all job openings are posted in the IMP, and open for application by staff with at least two years of service in their current positions | 32 positions (6.2% of all openings) were filled internally by eligible NEA staff |
| Educational Upgrading | Supports staff in their pursuit of higher education, with opportunities for application of sponsorship even after their course had commenced. This encourages staff to continuously upgrade, upskill and improve themselves through lifelong learning | 14 pursued higher qualifications through NEA's scholarship and sponsorship schemes |

Programmes that build technical capabilities and competencies, such as training workshops and specialist courses, are regularly conducted to help employees stay ahead in a dynamic operating environment. Some of these programmes have been jointly developed with industry experts and training providers.

NEA also conducted in-house programmes to build awareness on the future of work and encouraged employees to discover their career values and interests, and to have career conversations with their supervisors.

## DEVELOPING STAFF COMPETENCIES

NEA recognises the role of training and development in driving organisational productivity and growth. Our continued efforts in training and development ensure that employees will be equipped with the right skills and knowledge, to discharge their duties effectively and be able to meet future challenges. NEA's Singapore Environment Institute conducts competency-based training to build a relevant, resourceful and resilient workforce.

The Job Competency Matrix (JCM) framework forms the foundation of NEA's competency development and training. Under the framework, Skills Maps capture competencies needed for a job role, and Learning Intervention Matrices curate a list of recommended learning interventions for staff to plug competency gaps.

An annual Learning Needs Analysis is carried out to identify competency gaps and training, with 71 per cent of NEA workforce able to formulate their Departmental Learning Plans based on the JCM framework in FY2021.

Besides competency-based training, NEA also encourages our staff to embrace continuous learning for lifelong employability through our learning policy. Budget is also set aside for staff to attend personal development courses for improvement and innovation.

## EMBARKING ON DIGITAL LEARNING

NEA embarked on a digital learning transformation journey since the pandemic in 2020. The use of digital tools for training has now become one of the mainstream approaches for training NEA's workforce. A set of instructional strategies and tools were curated to support this new learning approach. Pre-reading resources were expanded, and digital toolbox introduced to cater to activity-based learning. The programmes were well-received and achieved overall satisfaction ratings of over 90 per cent.

In August 2021, NEA onboarded the Civil Service College's LEARN multi-tenancy platform, which allows the hosting of NEA's e-learning content accessible only to our staff. To date, staff can access over 60 new modules providing bite-size learning from their mobile devices. Beyond foundation modules covering the work of frontline departments and induction module to prep new hires, new topics developed include organisational competencies in finance, intellectual property, human resources, procurement, and investigation skills.

### CARING FOR STAFF'S WELL-BEING

Guided by the Employee Engagement Framework, NEA supports staff's well-being through the pandemic with relevant and meaningful initiatives, to enable staff to continue contributing in a productive, committed and purposeful manner.

#### Nurturing a Caring Culture

NEA believes in developing an engaged and socially responsible workforce that makes a positive impact on the environment and community, and contributes to NEA's social sustainability goals. The Captain Green Cares Community was formed in 2020 to care for our staff, raise the level of staff volunteerism in NEA, and support the initiatives of organisations that look out for the less privileged.

Achievements

- Employee participation rate in the ComChest SHARE programme increased from 92 per cent in FY2020 to 95 per cent in FY2021, qualifying NEA for the SHARE Platinum Award
- More than $37,800 in donations were raised through the various philanthropic initiatives supported by NEA
- Across NEA, 269 staff volunteers clocked in 2,275 volunteer hours to support WOG and NEA initiatives
- NEA was awarded the Bronze Certificate of Appreciation and featured on the SG Cares Giving Week website, in recognition for NEA's philanthropic efforts and support for the event

#### Fostering Cohesion and a Healthy Lifestyle

To build a healthy and active workforce, with positive relationships with colleagues and family members, we created opportunities for our staff to engage in sports and social activities.

The NEA Sports and Social Committee (NEA SSC) was formed in 2020 to make NEA more than just a workplace for staff, where friendships can be forged through various sports and social activities. Despite the pandemic and the changing work arrangement, NEA SSC continued to introduce various initiatives virtually to ensure staff remain connected to each other and stay active while working from home or in office. Complementing NEA SSC's initiatives, free health screening and influenza vaccinations were organised and made available at clinics across Singapore, for the convenience of staff.

#### Embracing Mental Wellness

Good mental health is an important part of overall well-being. Due to the prolonged pandemic, our top concerns are keeping our staff safe and healthy, and ensuring their mental health is cared for. NEA's practices are aligned with the Tripartite Advisory to support mental well-being at the workplace, and reviewed regularly to keep pace with changing circumstances.

## WALKING THE TALK

**Championing sustainability starts from within. NEA was the first public agency in Singapore to publish a sustainability report, and will continue to be one of the lead agencies for the GreenGov.SG initiative.**

### GUIDING NEA'S SUSTAINABILITY JOURNEY

NEA's core values are integral to our sustainability framework, which is shaped by four guiding principles.




W.I.S.E. supports NEA's key corporate sustainability pillars. Our desired outcomes are:

#### Engagement

- To achieve a high level of public trust
- For all staff to be NEA ambassadors

#### Environmental Sustainability Pillar

- To achieve a resource efficient office
- To achieve a clean and eco-friendly working environment

### Financial Sustainability Pillar

- To ensure responsible procurement, investments and operations

### Social Sustainability Pillar

- To have good workplace health and safety
- To have an expert and professional workforce
- To achieve strong corporate social responsibility

## CHAMPIONING SUSTAINABILITY THROUGH GOVERNANCE

A robust sustainability governance structure is set up to drive sustainability within NEA, with senior management responsible for setting the sustainability direction and endorsing the sustainability initiatives.

**SENIOR MANAGEMENT SUPPORT**

**NEA Management**
To provide overall leadership and direction on sustainability efforts in NEA
*Chair: CEO*
*Committee Members: Senior Management*

**MIDDLE MANAGEMENT SUPPORT**

**Sustainability Committee**
To discuss and track environmental sustainability efforts, and review NEA's resource consumption trend
*Chair: Group Director, Resource & Sustainability*
*Committee Members (Sustainability Champions): Directors*

**Sustainability Champions**
To help drive sustainable changes within their divisions, departments, and off-site premises

**STAFF INVOLVEMENT**

**Eco-office Champions**
To drive environmental sustainability

All NEA directors are Sustainability Champions, meeting quarterly at the Sustainability Committee meetings to discuss and track environmental sustainability efforts, and review NEA's resource consumption trend. These meetings are also attended by Eco-office Champions – working level staff who are nominated on a rotational basis to encourage new ideas and provide more opportunities for staff to be involved in NEA's sustainability journey.

## ACHIEVING WOG SUSTAINABILITY TARGETS

Building on efforts taken under the previous Public Sector Taking the Lead in Environmental Sustainability initiative, the new GreenGov.SG initiative was launched in July 2021 as a key enabler of the Singapore Green Plan 2030. Under GreenGov.SG, the public sector has set new and more ambitious sustainability targets that include peaking its carbon emissions around 2025, reducing its energy and water consumption by 10 per cent by 2030 from the average of 2018 to 2020 levels, and reducing the amount of waste disposed of by 30 per cent by 2030 from 2022 levels. To achieve these GreenGov.SG targets, NEA continues to implement resource management measures for our facilities, such as office buildings, laboratories, and hawker centres.

## ACHIEVING THE GREEN MARK

Under GreenGov.SG, all new public sector premises are to achieve at least Green Mark (GM) Platinum Super Low Energy (SLE) standards or equivalent, while all existing public sector premises are to achieve the GM Platinum SLE standards or equivalent upon the next major retrofitting. NEA has exceeded the minimum GM targets for many of our premises. These include Tiong Bahru Market and Choa Chu Kang Columbarium, which have attained the GM Platinum SLE and GM Platinum Positive Energy respectively due to measures such as good natural ventilation, energy-efficient lighting installation, and generation of on-site solar energy. Another 11 NEA premises, such as Marsiling Mall, Pasir Ris Central Hawker Centre, and One Punggol Hawker Centre, have also attained GM Gold and above standards. NEA will strive to meet the more ambitious GM requirements when these buildings are due for retrofit works.

## REDUCING CORPORATE CARBON FOOTPRINT

Although waste incineration is the main contributor to NEA's corporate carbon footprint, our Scope 1 emissions decreased by about 20 per cent compared to FY2020. This was attributed mainly to the scaling down of operations at Tuas Incineration Plant

as it moved towards the eventual cessation of operations in February 2022. On the other hand, our Scope 2 carbon emissions remained comparable to FY2020.

We continue to encourage staff to take public transport or carpool where possible, to manage our Scope 3 emissions. Due to the COVID-19 pandemic, the carbon emissions arising from overseas business travel in FY2021 remained low, due to the lower frequency of international events and official business visits as compared to pre-pandemic period. Local business travel remained relatively stable.

**Corporate Carbon Footprint[14]**

| | FY19 | | FY20 | | FY21 | |
|---|---|---|---|---|---|---|
| | Tonnes of $CO_2$ (% of total aggregate) | Ranking | Tonnes of $CO_2$ (% of total aggregate) | Ranking | Tonnes of $CO_2$ (% of total aggregate) | Ranking |
| **Direct Carbon Emissions (Scope 1)** | | | | | | |
| Waste Incineration[15] | 868,800 (99.2%) | 1 | 791,434 (99.2%) | 1 | 629,170 (99.0%) | 1 |
| NEA-owned Vehicles[16] | 4,861 (0.6%) | 2 | 4,006 (0.5%) | 2 | 3,996 (0.6%) | 2 |
| Liquefied Petroleum Gas (LPG) and Town Gas[17] | 2,437 (0.3%) | 4 | 2,637 (0.3%) | 4 | 2,316 (0.4%) | 4 |
| **Sub-total $CO_2$ Emissions (Scope 1)** | **876,098 tonnes $CO_2$** | | **798,077 tonnes $CO_2$** | | **635,482 tonnes $CO_2$** | |
| **Indirect Carbon Emissions (Scope 2)** | | | | | | |
| Electricity[18] (NEA Office Premises[19]) | 2,982 (65.7%) | 3 | 2,844 (66.1%) | 3 | 2,936 (66.4%) | 3 |
| Electricity[20] (NEA Installations[21]) | 1,556 (34.3%) | 5 | 1,458 (33.9%) | 5 | 1,489 (33.6%) | 5 |
| **Sub-total $CO_2$ Emissions (Scope 2)** | **4,539 tonnes $CO_2$** | | **4,302 tonnes $CO_2$** | | **4,425 tonnes $CO_2$** | |
| **Indirect Carbon Emissions (Scope 3)** | | | | | | |
| Local[22] Staff Business Travel | 327 (31.9%) | 7 | 302 (94.0%) | 6 | 277 (85.0%) | 6 |
| Overseas[23] Business Travel (Flight only) | 699 (68.1%) | 6 | 19 (6.0%) | 7 | 49 (15.0%) | 7 |
| **Sub-total $CO_2$ Emissions (Scope 3)** | **1,026 tonnes $CO_2$** | | **321 tonnes $CO_2$** | | **326 tonnes $CO_2$** | |

[14] *NEA's corporate carbon footprint is calculated in accordance with the GHG Protocol Corporate Accounting and Reporting Standard set by the World Business Council for Sustainable Development and World Resources Institute. This is in line with the 2006 Intergovernmental Panel on Climate Change (IPCC) Guidelines for National GHG Inventories, which is used in Singapore's international reporting to the UN Framework Convention on Climate Change. The computations for GHG emissions from waste incineration at the Tuas Incineration Plant and Tuas South Incineration Plant are also in accordance with the 2006 IPCC Guidelines for National GHG Inventories*

[15] *Consumption at incineration plants is largely dependent on public demand for NEA's services*

[16] *Carbon emissions for NEA-owned vehicles are calculated based on amount of fuel (diesel and petrol) used. Diesel consumed by vehicles, equipment and marine vessels used at Tuas Marine Transfer Station and Semakau Landfill were also included in the calculation of carbon emissions*

[17] *Carbon emissions are calculated for LPG and Town Gas used at Choa Chu Kang Crematorium and Mandai Crematorium respectively. The emissions are impacted by factors beyond NEA's control, such as the number of after-death services required*

[18] *Grid-average emissions factor data is used to calculate GHG emissions from electricity NEA purchased from the national grid. FY2021 figures are calculated based on 2020 electricity grid emission factors: 0.4080 kg $CO_2$ /kWh (www.ema.gov.sg/cmsmedia/Publications_and_Statistics/Statistics/18RSU.pdf)*

[19] *The Scope 2 emissions for NEA office premises have been restated for FY2019 and FY2020 to 2,982 tonnes $CO_2$ (from 2,000 tonnes $CO_2$) and 2,844 tonnes $CO_2$ (from 1,449 tonnes $CO_2$) respectively to include the carbon emissions from data centres, laboratories, and the meteorological outstations at Changi and Seletar*

[20] *The electricity consumed at Tuas South Incineration Plant, Tuas Incineration Plant and Tuas Marine Transfer Station was excluded from the carbon emissions calculation, as the electricity used at these premises is self-generated. For Semakau Landfill, the electricity consumed was generated by diesel generators on the landfill. The diesel consumed by generators had been accounted for under Scope 1 – NEA-owned vehicles*

[21] *The Scope 2 emissions for NEA installations have been restated for FY2019 and FY2020 to 1,556 tonnes $CO_2$ (from 1,545 tonne $CO_2$) and 1,458 tonnes $CO_2$ (from 1,364 tonnes $CO_2$) respectively to include the carbon emissions from Yishun Columbarium*

[22] *Carbon emissions were calculated based on local business travel by staff vehicles, excluding the travel journeys of NEA staff to and from their homes. According to the Land Transport Authority, cars and motorcycles are fuelled by petrol with average motor vehicle fuel efficiency of 10.45 L/100 km ('Singapore Land Transport Statistics in Brief', 2009)*

[23] *Carbon emissions arising from overseas business travel (flight only) are based on VITAL's Overseas Travel Report*

**Comparison of NEA's Corporate Emissions Intensity**

| | FY19 | FY20 | FY21 |
|---|---|---|---|
| Indirect Carbon Emissions (Scope 2+3)[24] (tonnes $CO_2$) | 5,565 | 4,623 | 4,751 |
| Occupants[25] | 3,783 | 1,926 | 2,312 |
| Carbon Emissions (tonnes $CO_2$ per occupant)[26] | 1.5 | 2.4 | 2.1 |

The amount of LPG used at Choa Chu Kang Crematorium and Town Gas used at Mandai Crematorium, are accounted for as part of NEA's corporate carbon footprint, together with diesel and petrol used in NEA-owned vehicles for government owned and operated after-death facilities. Total fuel consumption for FY2021 is 106 TJ and has remained comparable to the consumption in the last two FYs.

**Fuel Types Used**

| | FY19 | FY20 | FY21 |
|---|---|---|---|
| LPG and Town Gas | 43 TJ | 47 TJ | 52 TJ |
| Diesel and Petrol | 65 TJ | 54 TJ | 54 TJ |
| Total Fuel Consumption | 108 TJ | 101 TJ | 106 TJ |

## CONSERVING RESOURCES WITHIN NEA

NEA tracks our usage of electricity, water and paper, and encourages staff to reduce consumption of resources by adopting sustainability measures and practices. With the new set of sustainability targets under GreenGov.SG, NEA tracks the resource consumption using a new set of indices.

### Electricity

To align with GreenGov.SG reporting requirements, the electricity consumption at our offices and installations[27] are tracked using the following set of indicators:

- Electricity used per floor area
- Total electricity consumption

In FY2021, NEA's installations accounted for approximately 94.7 per cent of NEA's total electricity consumption. The electricity consumption at the installations decreased by 17.4 per cent as compared to FY2020. This was attributed mainly to the scaling down of operations at Tuas Incineration Plant as it moved towards the eventual cessation of operations in February 2022. On the other hand, the electricity consumption at offices in FY2021 remained comparable to that of FY2020.

Notwithstanding the COVID-19 pandemic period, NEA continued to implement energy efficiency measures in FY2021 to conserve resources internally. These include replacement of fluorescent lightings with LED lightings at a few hawker centres, installation of more energy-efficient air conditioners at our Central Regional Office, deployment of timers on water dispenser and air purifiers to reduce the duration of use, and replacement of old freezers with more efficient ones at the Environmental Health Institute's (EHI) laboratories. These measures achieved an estimated 60 MWh in energy savings annually.

**Electricity[28] Consumed at NEA Offices[29]**

| | FY19 | FY20 | FY21 |
|---|---|---|---|
| Electricity usage (MWh) | 7,091[30] | 6,961 | 7,196 |
| Electricity usage per unit floor area (MWh/$m^2$) | 0.2 | 0.2 | 0.2 |

**Electricity Consumed at NEA Installations[31]**

| | FY19 | FY20 | FY21 |
|---|---|---|---|
| Electricity usage (MWh) | 164,794 | 156,258 | 129,081 |
| Electricity usage per unit floor area (MWh/$m^2$) | 0.6 | 0.6 | 0.5 |

[24] *Only carbon emissions from Scope 2 and 3 are included as NEA has greater control over the business processes reported under these two scopes. Scope 1 carbon emissions are highly dependent on the demand for NEA services, such as the number of after-death services required*
[25] *Defined as the number of occupants, including the number of visitors, contractors and sub-contractors who had visited NEA's premises*
[26] *Carbon emissions intensity is derived as indirect carbon emissions (Scopes 2 and 3) per occupant*
[27] *NEA's installations include Choa Chu Kang Crematorium, Columbarium and Cemetery, Mandai Crematorium and Columbarium, Yishun Columbarium, Kranji State Cemetery, Semakau Landfill, Tuas Marine Transfer Station, Tuas Incineration Plant and Tuas South Incineration Plant. Consumption at NEA installations is largely dependent on public demand for NEA's services*
[28] *This refers to the electricity obtained from the national power grid*
[29] *To align with the scope of the GreenGov.SG initiative, the electricity consumption figures for NEA offices in FY2019 and FY2020 have been restated to 7,091 MWh (from 4,776 MWh) and 6,961 MWh (from 3,548 MWh) respectively to include the electricity consumption at data centres, laboratories, and meteorological outstations at Changi and Seletar*
[30] *The electricity consumed at two former NEA premises is not included in this figure as these two premises had been vacated in mid FY2019*
[31] *The electricity consumption figures for NEA installations for FY2019 and FY2020 have been restated to 164,794 MWh (from 164,734 MWh) and 156,258 MWh (from 156,003 MWh) respectively to include the electricity consumption at Yishun Columbarium. The electricity consumption at Kranji State Cemetery was not included, as the energy consumption at this premises is attributed to street lights that are powered by solar panels*

Aside from drawing electricity from the grid, NEA's premises also make use of solar energy and electricity generated from waste incineration. In FY2021, the total electricity generated from solar and waste-to-energy incineration process was 546,588 MWh. The electricity generated is equivalent to the average annual electricity consumed by about 130,000[32] Singapore households.

Currently, eight NEA-managed premises have solar panels installed as part of the WOG SolarNova programme. The Centre for Climate Research Singapore and Semakau Landfill are also installed with solar capacity. These 10 premises generated 1,418 MWh of electricity in FY2021.

Electricity generated from waste incineration adds to NEA's pool of energy sources, powering the operations in the incineration plants and Tuas Marine Transfer Station. Any excess energy from the waste-to-energy incineration process is sold to the grid. The amount sold to the grid in FY2021 was 422,036 MWh.

### Water

The water consumption at our offices and installations are tracked using the two indicators:

- Water used per occupant per day
- Total water consumption

The water consumption at NEA's installations increased by 16.3 per cent in FY2021, accounting for about 79.6 per cent of our total water usage. This was partly due to the extensive washing of buildings prior to repair and redecoration works at Tuas Marine Transfer Station, as well as increased washing and cleaning at Choa Chu Kang Columbarium caused by algae growth. There was also a case of water leakage from a burst underground water pipeline, which was repaired promptly.

To reduce water usage, we have replaced our washing machines at the EHI laboratories with 4-tick Water Efficiency Labelling Scheme (WELS) rating models, and maximise the load per wash. We will continue to replace water fittings with 3-tick WELS rating models through repair and redecoration works. Regular monitoring and reporting of water usage will also be conducted so that any spike in consumption can be investigated and rectified quickly.

Water is reused as much as possible across NEA's installations. For instance, at the Tuas Incineration Plant, water is drained from its boiler for use in general cleaning, such as washing the reception hall and quenching the bottom ash at the wet ash extractor.

**PUB Water Consumed at NEA Offices[33]**

| | FY19 | FY20 | FY21 |
|---|---|---|---|
| Water consumed (L) | 26,190,000 | 19,259,000 | 18,487,000 |
| Water consumed per occupant per day (L/person/day) | 37.6 | 64.0 | 53.0 |

**PUB Water Consumed at NEA Installations[34]**

| | FY19 | FY20 | FY21 |
|---|---|---|---|
| Water consumed (L) | 63,266,000 | 62,166,000 | 72,270,000 |
| Water consumed per occupant per day (L/person/day) | 104.3 | 122.9 | 122.9 |

NEA uses two alternative water sources other than PUB's potable water at NEA's installations – NEWater and rainwater.

NEA's Incineration Plants consumed 457,464,000 L of NEWater in FY2021, for purposes such as general cleaning, cooling hot ash, suppressing dust, and producing demineralised water for boiler usage.

As part of NEA's water conservation efforts, NEA sets up rain harvesting systems at several NEA premises. The rainwater collected is used to irrigate the grounds and to clean and run NEA's operations, such as the cooling of hot ash and suppression of dust at incineration plants and Tuas Marine Transfer Station. Tuas South Incineration Plant and Tuas Marine Transfer Station are equipped with meters to track the amount of

[32] *Based on 2019 average monthly electricity consumption for four-room households of 359.1 KWh/month (www.ema.gov.sg/cmsmedia/8RSU.xlsx)*

[33] *To align with the scope of GreenGov.SG initiative, the water consumption figures for NEA offices for FY2019 and FY2020 have been restated to 26,190,000 L (from 25,428,000 L) and 19,259,000 L (from 17,208,000 L) respectively to include the water consumption at laboratories, and meteorological outstations at Changi and Seletar*

[34] *The water consumption figures for NEA installations for FY2019 and FY2020 have been restated to 63,266,000 L (from 67,056,000 L) and 62,166,000 L (from 65,563,000 L) respectively to include the water consumption at Yishun Columbarium and Kranji State Cemetery*

rainwater used. The total volume of rainwater collected in FY2021 was 33,679,000 L, equivalent to the average amount of water consumed by about 654[35] Singapore residents in their homes annually.

### Waste

With limited land space for waste disposal, NEA is working to extend the lifespan of Singapore's only landfill by reducing our waste generation and increasing recycling.

The bulk of the waste generated comes from the offices[36]. In FY2021, 34 tonnes of non-recyclable waste were disposed of, similar to FY2020. To encourage staff to recycle, there are recycling bins in printing rooms and pantries at NEA Headquarters and in convenient locations at off-site premises. There are recycling bins for toner cartridges and e-waste at NEA Headquarters. In FY2021, NEA collected 2,098 kg[37] recyclables consisting of paper (73.8 per cent), plastics (23.8 per cent) and metals (2.4 per cent).

To reduce wastage, we encourage staff to reuse furniture and recycle unwanted items when shifting to a new office. We also encourage staff to share or view documents digitally, and to print only when necessary. For example, NEA adopted digital business cards in place of printed business cards, which saved paper and cost. Since August 2014, NEA has progressively replaced our existing printers with new printers that come with the print-count feature. This feature has enabled us to identify paper-intensive processes and look into digitalising some of these processes[38].

### OVERSEEING BIOHAZARDOUS WASTE DISPOSAL

In FY2021, 64,320 L of biohazardous waste was generated from EHI's laboratory work. While this could not be recycled, NEA has put in place strict disposal processes to prevent contaminating the environment. The biohazardous waste was stored in purpose-built waste bins before collection by licensed biohazardous waste collectors – Asia Medical Enviro Services Pte Ltd in FY2021. The biohazardous waste was then transported to a dedicated incinerator for incineration.

[35] *Based on Singapore's annual household water consumption per capita of 141 L per day in 2019 (www.pub.gov.sg/watersupply/singaporewaterstory)*

[36] *Waste data is compiled from five premises – Singapore Environment Institute at Lifelong Learning Institute, Choa Chu Kang Crematorium, Columbarium and Cemetery, Western Regional Office at Albert Winsemius Lane, Semakau Landfill and Tuas Marine Transfer Station*

[37] *The amount of recyclables collected in FY2021 is significantly lesser than FY2020 due to a one-off recycling event of 3,910 kg of shredded paper in FY2020 at one of NEA's off-site premises*

[38] *The print count data for FY2021 is not reported due to replacement of printers at most of our premises between April and December 2021*

# About This Report

In this Integrated Sustainability Report, NEA provides an account of our corporate, financial and sustainability performance in FY2021 (1 April 2021 to 31 March 2022).

We hope that this report will serve as a useful reference for other organisations and agencies, and inspire them to embark on their own journey of sustainability.

## STANDARDS IMPLEMENTED

This Report has been prepared in accordance with:

- The GRI Standards – Core option with submission to the GRI for the Materiality Disclosures Service. GRI Services reviewed that the GRI content index is clearly presented, and the references for Disclosures 102-40 to 102-49 align with appropriate sections in the body of the report.
- The Integrated Reporting Framework set by the International Integrated Reporting Council, maintained under the International Financial Reporting Standards Foundation.

## REPORTING SCOPE

This report documents organisational activities, achievements and financial performance across NEA's 22 key premises, unless otherwise stated.

In FY2021, there are five stand-alone premises, nine tenanted office premises and eight industrial premises.

### Stand-alone Office Premises

i. Centre for Climate Research Singapore at Kim Chuan Road
ii. NEA Regional Offices:
    - Eastern Regional Office at Tannery Lane
    - Eastern Regional Office (Satellite Office) at Sin Ming Drive
    - Central Regional Office at Jalan Bukit Merah
    - Western Regional Office at Albert Winsemius Lane

### Tenanted Office Premises

i. Environmental Health Institute at Biopolis
ii. Environmental Health Institute at Ang Mo Kio Tech Place II
iii. Hawker Centres Group at HDB Hub
iv. Meteorological Service Singapore at Changi Airport
v. National Radiochemistry Laboratory at CREATE
vi. NEA Headquarters at Environment Building
vii. Secondary Standards Dosimetry Laboratory at Health Sciences Authority
viii. Singapore Environment Institute at Lifelong Learning Institute
ix. Pollution Control Division 1 and 2 at Bukit Merah Central

### Industrial Premises

i. Choa Chu Kang Crematorium, Columbarium and Cemetery
ii. Mandai Crematorium and Columbarium
iii. Yishun Columbarium
iv. Kranji State Cemetery
v. Semakau Landfill
vi. Tuas Marine Transfer Station
vii. Tuas South Incineration Plant
viii. Tuas Incineration Plant

## ASSURANCE

The data included in this report had undergone rigorous internal processes such as data monitoring and verification to ensure accuracy. While data and information were reviewed by external sustainability consultant, we have not sought external assurance for the reporting period.

## ENQUIRIES

As part of NEA's environmental conservation efforts, the report will not be printed. Digital copies of past and current reports can be found on NEA's corporate website: www.nea.gov.sg

For enquiries, please contact NEA's Corporate Communications Division:

Ms Eleanor Goh eleanor_goh@nea.gov.sg

# INTEGRATED REPORTING APPROACH

NEA's CEO acknowledges his responsibility to ensure the integrity of this report, which has been prepared with reference to the International Integrated Reporting Framework. Integrated Reporting is a process founded on integrated thinking that results in a report demonstrating how we create value over time with our strategy, governance, performance and prospects.

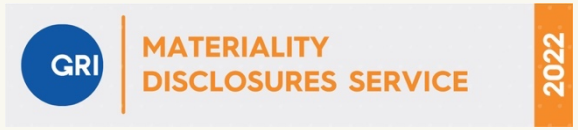
We use the Content Elements and Principles of Integrated Reporting as a guide to improve our internal planning and external reporting. For the purpose of this report, NEA defines the following timeframes:

- Short-term: 1 to 2 years
- Medium-term: 2 to 5 years
- Long-term: 5 years and beyond

**Integrated Reporting Index**

| Content Elements | Section Reference |
|---|---|
| A. Organisational overview and external environment | Vision, Mission, Roles and Core Values<br>About the National Environment Agency |
| B. Governance | Corporate Governance<br>Walking the Talk |
| C. Business model | Value Creation Model |
| D. Risks and opportunities | Material Issues<br>Risks and Opportunities |
| E. Strategy and resource allocation | Chairman's Foreword<br>CEO's Message<br>Value Creation Model |
| F. Performance | Value Creation Model<br>Ensuring a Clean and Sustainable Environment for Singapore<br>Together with Our Partners and the Community<br>Enabled by a High Performance and Future-ready NEA |
| G. Outlook | Chairman's Foreword<br>CEO's Message |
| H. Basis of preparation and presentation | Material Issues<br>About This Report |

# GRI Content Index




For the Materiality Disclosures Service, GRI Services reviewed that the GRI content index is clearly presented and the references for Disclosures 102-40 to 102-49 align with appropriate sections in the body of the report.

| GRI Standard Number | Disclosure Number | Section Titles | Page No. | Direct Answers / Reasons for Omission |
|---|---|---|---|---|
| **GRI 101: Foundation 2016** | | | | |
| **General Disclosures** | | | | |
| **Organisation Profile** | | | | |
| **GRI 102 : General Disclosures 2016** | 102-1 Name of the organisation | About the National Environment Agency | 9 | |
| | 102-2 Activities, brands, products, and services | About the National Environment Agency | 9 - 10 | |
| | 102-3 Location of headquarters | About the National Environment Agency<br>About This Report | 9<br>49 | |
| | 102-4 Location of operations | About the National Environment Agency<br>About This Report | 9<br>49 | |
| | 102-5 Ownership and legal form | About the National Environment Agency | 9 | |
| | 102-6 Markets served | About the National Environment Agency | 9 | |
| | 102-7 Scale of the organisation | About the National Environment Agency<br>Strong Corporate Practices for a Sustainable Organisation | 9<br>38 | |
| | 102-8 Information on employees and other workers | Strong Corporate Practices for a Sustainable Organisation | 38 | A significant portion of NEA's activities are performed by workers who are not NEA employees. These activities include waste collection, public cleaning services, pest control services, public hygiene enforcement, and call-centre services. |
| | 102-9 Supply chain | Strong Corporate Practices for a Sustainable Organisation | 40 - 41 | |
| | 102-10 Significant changes to the organisation and its supply chain | CEO's Message | 4 | |
| | 102-11 Precautionary Principle or approach | Risks and Opportunities | 15 - 16 | |
| | 102-12 External initiatives | Public Sector Taking the Lead in Environmental Sustainability | 32 - 33 | NEA aligns our HR practices with the Tripartite Guidelines on Fair Employment Practices. |
| | 102-13 Membership of associations | About the National Environment Agency | 9 | |

| | | | | |
|---|---|---|---|---|
| **Strategy** | | | | |
| **GRI 102 : General Disclosures 2016** | 102-14 Statement from senior decision-maker | Chairman's Foreword<br>CEO's Message | 4<br>5 | |
| | 102-15 Key impacts, risks, and opportunities | Material Issues<br>Risks and Opportunities | 11 - 14<br>15 - 17 | |
| **Ethics and Integrity** | | | | |
| **GRI 102 : General Disclosures 2016** | 102-16 Values, principles, standards, and norms of behaviour | Vision, Mission, Roles and Core Values | 3 | |
| **Governance** | | | | |
| **GRI 102 : General Disclosures 2016** | 102-18 Governance structure | Corporate Governance<br>Walking the Talk | 6<br>44 | |
| **GRI 419: Socioeconomic Compliance 2016** | 419-1 Non-compliance with laws and regulations in the social and economic area | - | - | In FY2021, there were no incidents of social or economic non-compliance, such as discrimination and corruption. |
| **Stakeholder Engagement** | | | | |
| **GRI 102 : General Disclosures 2016** | 102-40 List of stakeholder groups | Material Issues | 11 | |
| | 102-41 Collective bargaining agreements | - | | 94% of staff are eligible to join the Union and are covered under collective agreements. |
| | 102-42 Identifying and selecting stakeholders | Material Issues | 11 | |
| | 102-43 Approach to stakeholder engagement | Environmental Services Industry as a Force Multiplier for Mission Success<br>Active Community as Stewards of the Environment<br>Public Sector Taking the Lead in Environmental Sustainability<br>International Partnerships as Strategic Levers | 31<br>32<br>33<br>34 | No engagement was undertaken specifically as part of the report preparation process. |
| | 102-44 Key topics and concerns raised | Environmental Services Industry as a Force Multiplier for Mission Success<br>Active Community as Stewards of the Environment<br>Public Sector Taking the Lead in Environmental Sustainability<br>International Partnerships as Strategic Levers | 31<br>32<br>33<br>34 | |
| **Reporting Practice** | | | | |
| **GRI 102 : General Disclosures 2016** | 102-45 Entities included in the consolidated financial statements | About This Report | 49 | |
| | 102-46 Defining report content and topic Boundaries | Material Issues | 11 - 14 | |
| | 102-47 List of material topics | Material Issues | 11 - 14 | |

| | | | | |
|---|---|---|---|---|
| | 102-48 Restatements of information | - | | The resource consumption data and carbon footprint figures have been restated for FY2019 and FY2020 to align with the new GreenGov.SG initiative. |
| | 102-49 Changes in reporting | - | | There were no significant changes. |
| | 102-50 Reporting period | About This Report | 49 | |
| | 102-51 Date of most recent report | - | | The previous report was published in September 2021. |
| | 102-52 Reporting cycle | - | | This report will continue to be published on an annual basis. |
| | 102-53 Contact point for questions regarding the report | About This Report | 49 | |
| | 102-54 Claims of reporting in accordance with the GRI Standards | - | | This report has been prepared in accordance with the GRI Standards: Core option. |
| | 102-55 GRI content index | GRI Content Index | 51- 57 | |
| | 102-56 External assurance | - | | The data included in this report had undergone rigorous internal processes such as data monitoring and verification to ensure accuracy. While data and information were reviewed by external sustainability consultant, we have not sought external assurance for this reporting period. |
| **Topic Specific Disclosures** | | | | |
| **Economic Performance** | | | | |
| **GRI 103 : Management Approach 2016** | 103-1 Explanation of the material topic and its Boundary | Material Issues | 11 - 14 | |
| | 103-2 The management approach and its components | Strong Corporate Practices for a Sustainable Organisation | 40 - 41 | |
| | 103-3 Evaluation of the management approach | Strong Corporate Practices for a Sustainable Organisation | 40 - 41 | |
| **GRI 201 : Economic Performance 2016** | 201-1 Direct economic value generated and distributed | Refer to Annual Financial Statements<br>Strong Corporate Practices for a Sustainable Organisation | 43 | |
| | 201-2 Financial implications and other risks and opportunities due to climate change | Risks and Opportunities<br>Providing Timely and Reliable Weather and Climate Services | 16<br>25 - 26 | |
| | 201-3 Defined benefit plan obligations and other retirement plans | Refer to Annual Financial Statements, Defined Contribution Plan<br>Strong Corporate Practices for a Sustainable Organisation | 39 | |
| | 201-4 Financial assistance received from government | Strong Corporate Practices for a Sustainable Organisation | 41 | |

| | | | | |
|---|---|---|---|---|
| **Procurement Practices** | | | | |
| **GRI 103 : Management Approach 2016** | 103-1 Explanation of the material topic and its Boundary | Material Issues | 11 - 14 | |
| | 103-2 The management approach and its components | Strong Corporate Practices for a Sustainable Organisation | 40 - 41 | |
| | 103-3 Evaluation of the management approach | Strong Corporate Practices for a Sustainable Organisation | 40 - 41 | |
| **GRI 204 : Procurement Practices 2016** | 204-1 Proportion of spending on local suppliers | Strong Corporate Practices for a Sustainable Organisation | 41 | |
| **Energy** | | | | |
| **GRI 103 : Management Approach 2016** | 103-1 Explanation of the material topic and its Boundary | Material Issues<br>Risks and Opportunities<br>Sustainable and Resource Efficient Singapore | 11 - 14<br>15 - 17<br>23 - 25 | |
| | 103-2 The management approach and its components | Sustainable and Resource Efficient Singapore<br>Walking the Talk | 23 - 25<br>44 - 47 | |
| | 103-3 Evaluation of the management approach | Walking the Talk | 44 - 47 | |
| **GRI 302 : Energy 2016** | 302-1 Energy consumption within the organisation | Walking the Talk | 44 - 47 | NEA's total energy consumption (electricity and fuel) was 596.2 TJ during the reporting period. |
| | 302-2 Energy consumption outside of the organisation | - | | Information unavailable. NEA has not started tracking energy consumption outside of the organisation. |
| | 302-3 Energy intensity | Walking the Talk | 44 - 47 | |
| | 302-4 Reduction of energy consumption | Walking the Talk | 44 - 47 | |
| | 302-5 Reductions in energy requirements of products and services | - | | Not applicable to NEA. |
| **Water and Effluents** | | | | |
| **GRI 103 : Management Approach 2016** | 103-1 Explanation of the material topic and its Boundary | Material Issues<br>Safe, Healthy and Conducive Living Environment<br>Sustainable and Resource Efficient Singapore | 11 - 14<br>21 - 23<br>25 | |
| | 103-2 The management approach and its components | Safe, Healthy and Conducive Living Environment<br>Walking the Talk | 21 - 23<br>43 - 48 | |
| | 103-3 Evaluation of the management approach | Walking the Talk | 43 - 48 | |
| **GRI 303 : Water and Effluents 2018** | 303-1 Interactions with water as a shared resource | Safe, Healthy and Conducive Living Environment<br>Walking the Talk | 21 - 23<br>43 - 48 | No water bodies and related habitats were significantly affected by water withdrawal, discharges or runoff. |
| | 303-2 Management of water discharge-related impacts | Safe, Healthy and Conducive Living Environment | 22 | |

| | | | | |
|---|---|---|---|---|
| | 303-3 Water withdrawal | Walking the Talk | 47 | Water withdrawal is tracked using water meters. All water withdrawal are from third-party sources (PUB). |
| | 303-4 Water discharge | Safe, Healthy and Conducive Living Environment<br>Walking the Talk | 22 | |
| **Emissions** | | | | |
| **GRI 103 : Management Approach 2016** | 103-1 Explanation of the material topic and its Boundary | Material Issues<br>Safe, Healthy and Conducive Living Environment | 11 - 14<br>21 - 23 | |
| | 103-2 The management approach and its components | Safe, Healthy and Conducive Living Environment<br>Walking the Talk | 21 - 23<br>43 - 48 | |
| | 103-3 Evaluation of the management approach | Walking the Talk | 43 - 48 | |
| **GRI 305 : Emissions 2016** | 305-1 Direct (Scope 1) GHG emissions | Walking the Talk | 45 | |
| | 305-2 Energy indirect (Scope 2) GHG emissions | Walking the Talk | 45 | |
| | 305-3 Other indirect (Scope 3) GHG emissions | Walking the Talk | 45 | |
| | 305-4 GHG emissions intensity | Walking the Talk | 46 | |
| | 305-5 Reduction of GHG emissions | - | | Information is currently unavailable. |
| | 305-6 Emissions of ozone-depleted substances (ODS) | - | | NEA is not a significant emitter of ODS – this is not a material topic. |
| | 305-7 Nitrogen oxides ($NO_x$), sulfur oxides ($SO_x$), and other significant air emissions | Safe, Healthy and Conducive Living Environment | 21 - 23 | Under the Environmental Protection and Management (Air Impurities) Regulations, air emissions is reported based on $mg/Nm^3$. |
| **Waste** | | | | |
| **GRI 103 : Management Approach 2016** | 103-1 Explanation of the material topic and its Boundary | Material Issues<br>Sustainable and Resource Efficient Singapore | 11 - 14<br>23 - 25 | |
| | 103-2 The management approach and its components | Sustainable and Resource Efficient Singapore | 23 - 25 | |
| | 103-3 Evaluation of the management approach | Walking the Talk | 43 - 48 | |
| **GRI 306 : Waste 2020** | 306-1 Waste generation and significant waste-related impacts | Sustainable and Resource Efficient Singapore | 23 - 25 | |
| | 306-2 Management of significant waste-related impacts | Walking the Talk | 43 - 48 | NEA does not import or export hazardous waste. |
| | 306-3 Waste generated | Walking the Talk | 48 | |
| | 306-4 Waste diverted from disposal | Walking the Talk | 48 | |

| | | | | |
|---|---|---|---|---|
| **Environmental Compliance** | | | | |
| **GRI 103 : Management Approach 2016** | 103-1 Explanation of the material topic and its Boundary | Material Issues<br>Safe, Healthy and Conducive Living Environment | 11 - 14<br>21 - 23 | |
| | 103-2 The management approach and its components | Sustainable and Resource Efficient Singapore<br>Strong Capabilities for Mission Success | 21 - 23<br>35 - 36 | |
| | 103-3 Evaluation of the management approach | Safe, Healthy and Conducive Living Environment | 21 - 23 | |
| **GRI 307 : Environmental Compliance 2016** | 307-1 Non-compliance with environmental laws and regulations | - | | NEA did not receive any environmental fines and sanctions. |
| **Employment** | | | | |
| **GRI 103 : Management Approach 2016** | 103-1 Explanation of the material topic and its Boundary | Material Issues | 11 - 14 | |
| | 103-2 The management approach and its components | Strong Corporate Practices for a Sustainable Organisation | 36 - 38 | |
| | 103-3 Evaluation of the management approach | Strong Corporate Practices for a Sustainable Organisation | 36 - 38 | |
| **GRI 401 : Employment 2016** | 401-1 New employee hires and employee turnover | Strong Corporate Practices for a Sustainable Organisation | 38 | |
| | 401-2 Benefits provided to full-time employees that are not provided to temporary or part-time employees | Strong Corporate Practices for a Sustainable Organisation | 39 | |
| | 401-3 Parental leave | Strong Corporate Practices for a Sustainable Organisation | 39 | |
| **Occupational Health and Safety** | | | | |
| **GRI 103 : Management Approach 2016** | 103-1 Explanation of the material topic and its Boundary | Material Issues | 11 - 14 | |
| | 103-2 The management approach and its components | Strong Corporate Practices for a Sustainable Organisation | 36 - 40 | |
| | 103-3 Evaluation of the management approach | Strong Corporate Practices for a Sustainable Organisation | 36 - 40 | |
| **GRI 403 : Occupational Health and Safety 2018** | 403-1 Occupational health and safety management system | Strong Corporate Practices for a Sustainable Organisation<br>Walking the Talk | 36 - 40 | |
| | 403-2 Hazard identification, risk assessment, and incident investigation | Strong Corporate Practices for a Sustainable Organisation | 40 | |
| | 403-3 Occupational health services | Strong Corporate Practices for a Sustainable Organisation | 40 | |
| | 403-4 Worker participation, consultation, and communication on occupational health and safety | Strong Corporate Practices for a Sustainable Organisation | 40 | NEA has a Workplace Safety and Health (WSH) Driver and WSH Representatives from each department and off-site premises. All NEA staff are represented by the safety committee. |
| | 403-5 Worker training on occupational health and safety | Strong Corporate Practices for a Sustainable Organisation | 40 | |
| | 403-6 Promotion of worker health | Strong Corporate Practices for a Sustainable Organisation | 40 | |
| | 403-7 Prevention and mitigation of occupational health and safety impacts directly linked by business relationships | Strong Corporate Practices for a Sustainable Organisation | 40 | The safety of workers other than employees are also covered by our WSH policy and governance. |

| | | | | |
|---|---|---|---|---|
| | 403-9 Work-related injuries | Strong Corporate Practices for a Sustainable Organisation | 40 | NEA aligns with MOM in using injury rate per 100,000 workers to assess WSH performance. |
| **Training and Education** | | | | |
| **GRI 103 : Management Approach 2016** | 103-1 Explanation of the material topic and its Boundary | Material Issues<br>Conducive Culture for Collaboration, Learning and Innovation | 11 - 14<br>41 | |
| | 103-2 The management approach and its components | Conducive Culture for Collaboration, Learning and Innovation | 41 - 43 | |
| | 103-3 Evaluation of the management approach | Conducive Culture for Collaboration, Learning and Innovation | 41 - 43 | |
| **GRI 404: Training and Education 2016** | 404-2 Programs for upgrading employee skills and transition assistance programs | Conducive Culture for Collaboration, Learning and Innovation | 41 - 43 | NEA has pivoted towards competency-based learning to focus on both individual and organisational competency, rather than number of hours, hence NEA is not reporting training hours from FY21 onwards. |
| | 404-3 Percentage of employees receiving regular performance and career development reviews | Conducive Culture for Collaboration, Learning and Innovation | 41 - 42 | |




**National Environment Agency**

40 Scotts Road, #13-00
Environment Building
Singapore 228231

www.nea.gov.sg